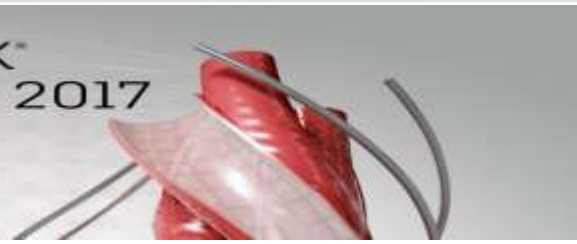


Lecture 1

مقدمة الى برنامج اوتوكاد

Dr. Ghalib Ahmed Salman.

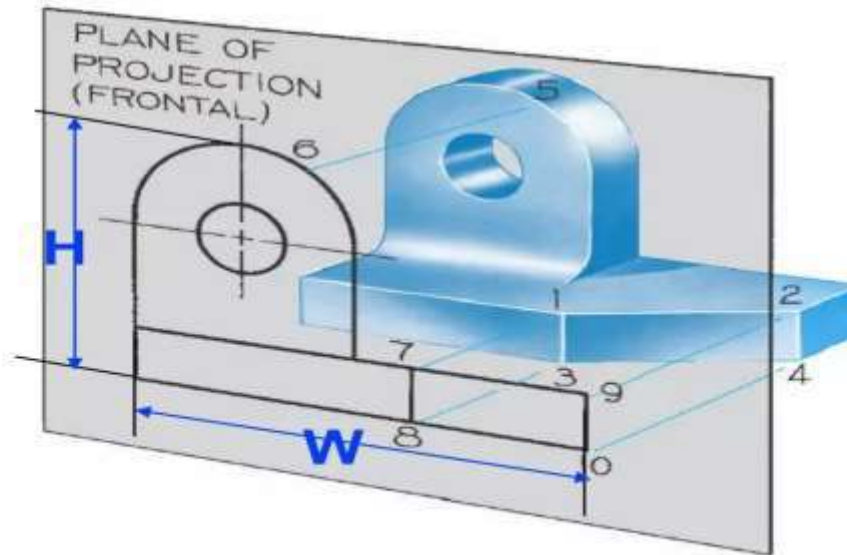
dr.ghalib@mtu.edu.iq



ماهو الرسم الهندسي؟

تعريف الرسم الهندسي

هو تمثيل لجزء أو مجموعة أجزاء من منتج ما بمساقط متعامدة أو مجسمة، مع إضافة الأبعاد والبيانات التوضيحية





ما هو برنامج AutoCAD؟

□ (**Auto** Computer **A**ided **D**esign)

□ هو برنامج للرسم والتصميم بمساعدة الحاسوب يدعم إنشاء الرسومات ثنائية وثلاثية الأبعاد.

□ تم تطوير هذا البرنامج منذ عام 1982 كتطبيق للحواسيب الشخصية، ومنذ عام 2010 أصبح متوفراً كتطبيق ويب يعمل خلال المتصفحات والهواتف الذكية يعتمد مبدأ **التخزين السحابي**



خصائص برنامج AutoCAD

□ يعمل الاتوكاد بنفس قواعد الرسم الهندسي الورقية لكنه يوفر الامكانيات التالية:

✓ السرعة

✓ الدقة

✓ امكانية التعديل

✓ سهولة الخزن بكميات كبيرة وفترات طويلة



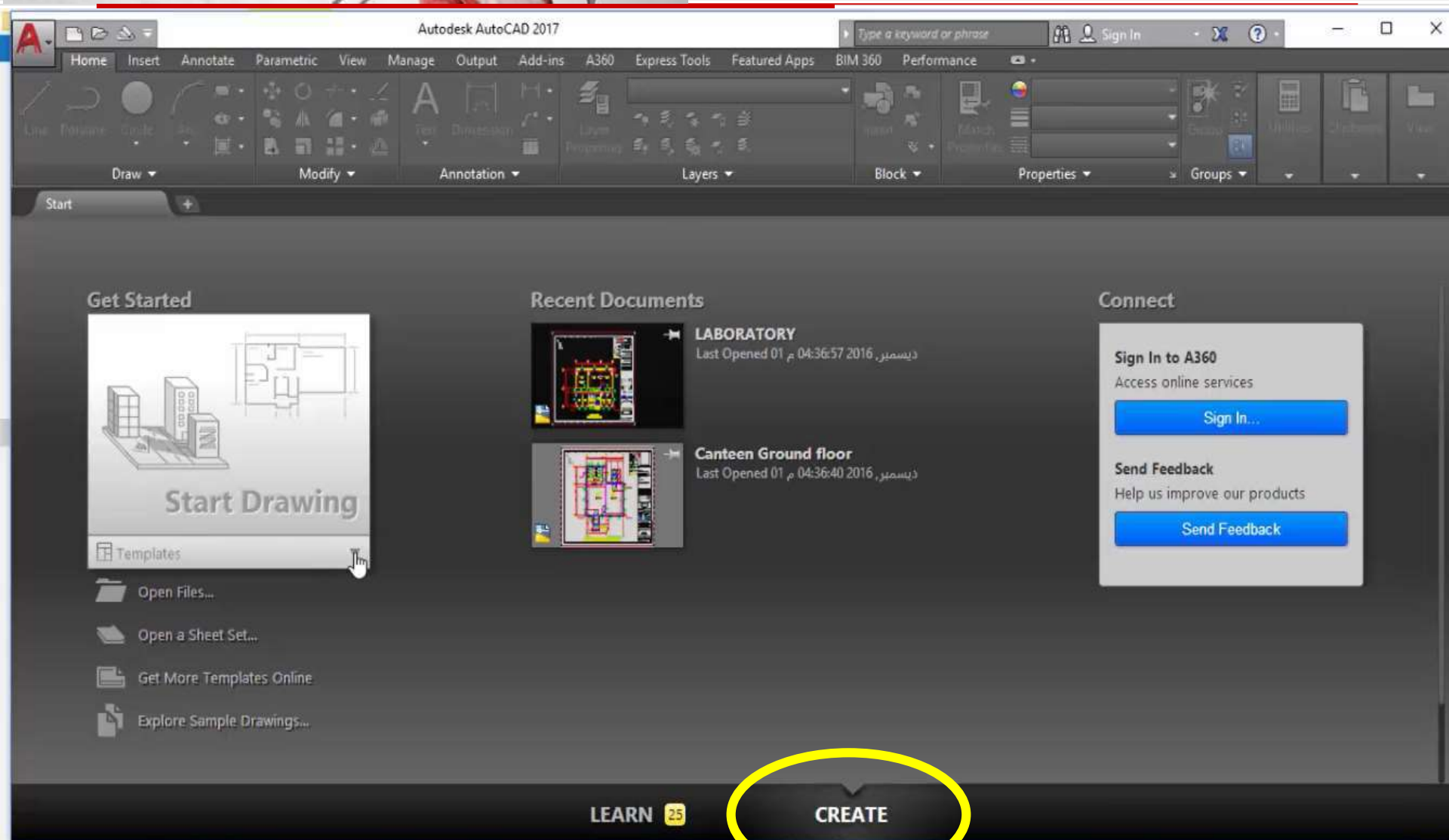
اهداف تعلم AutoCAD؟

- (1) ان يتعرف الطالب على أهمية الرسم الهندسي
- (2) ان يتعلم الطالب كيفية تخيل الأشكال الهندسية
- (3) ان يميز الطالب بين مكونات وأجزاء الشيء المرسوم
- (4) الرسم الهندسي هو لغة المهندسين

PAPER SIZE

Sl. No.	Designation	Trimmed Size in mm	Un Trimmed Size (Min) in mm
1	A0	841 x 1189	880 x 1230
2	A1	594 x 841	625 x 880
3	A2	420 x 594	450 x 625
4	A3	297 x 420	330 x 450
5	A4	210 x 297	240 x 330
6	A5	148 x 210	165 x 240

اجزاء واجهة AutoCAD؟



اجزاء واجهة AutoCAD؟

Autodesk AutoCAD 2017

Home Insert Annotate Parametric View Manage Output Add-ins A360 Express Tools Featured Apps BIM 360 Performance

Draw Modify Annotation Layers Block Properties Groups

Start

What's New

AutoCAD New Features Overview
See the key enhancements in Autodesk AutoCAD 2017.
4:14

Getting Started Videos

Tour the User Interface
View a demo of the basic tools in the user interface.
4:16

Create 2D Objects
Learn how to create and organize 2D object using layers, basic drawing tools, ortho, and object snap.
2:03

Modify 2D Objects

Feature Videos

Autodesk desktop app
The Autodesk desktop app delivers updates, security patches, and learning content to your desktop.
1:38

Learning Tips

For a more precise placement, attach a PDF before importing it. You can then use the ALIGN command to orient and place the PDF.
[More information...](#)

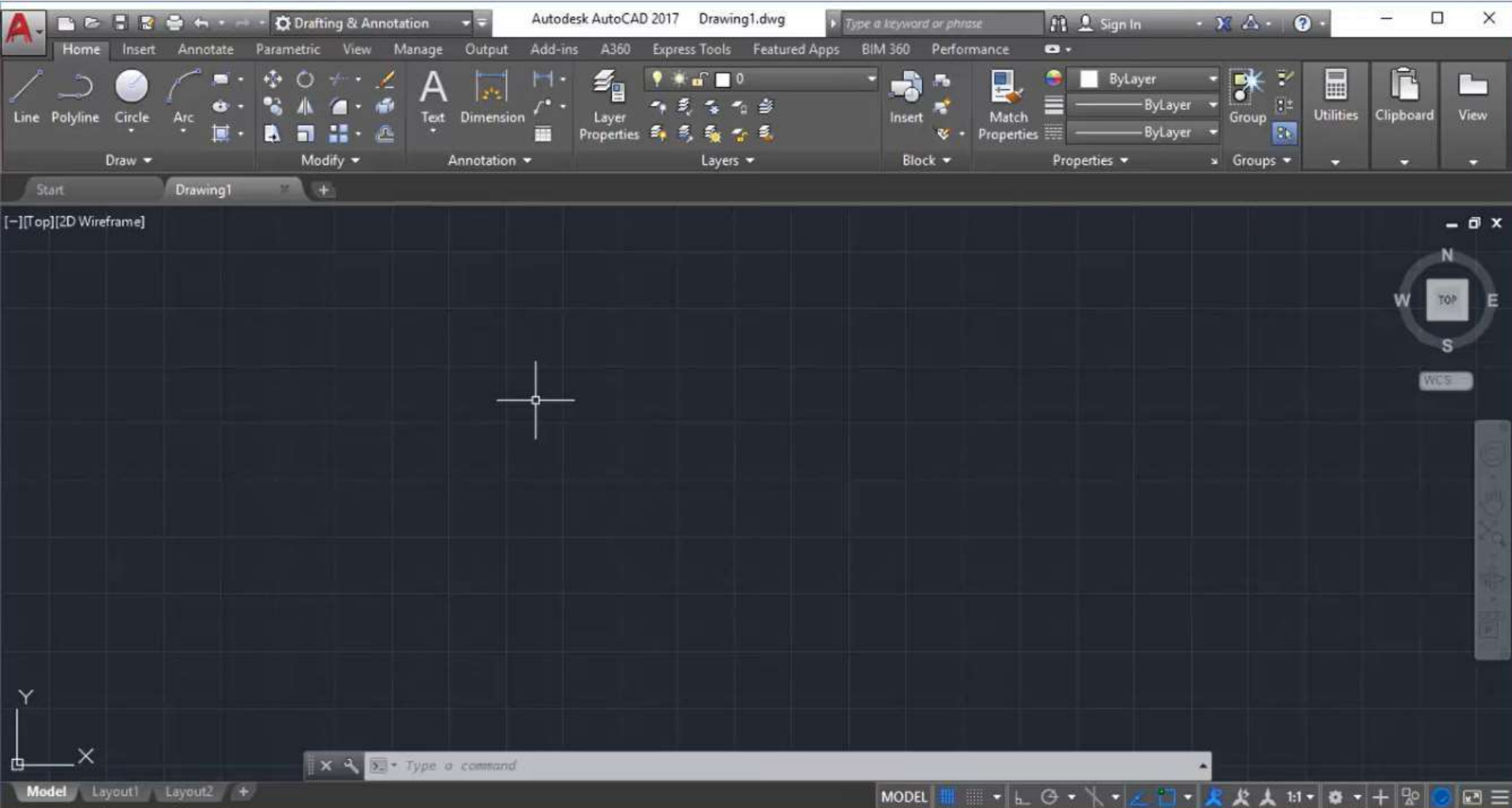
Online Resources

Explore training materials on the web to sharpen your skills or learn something new.

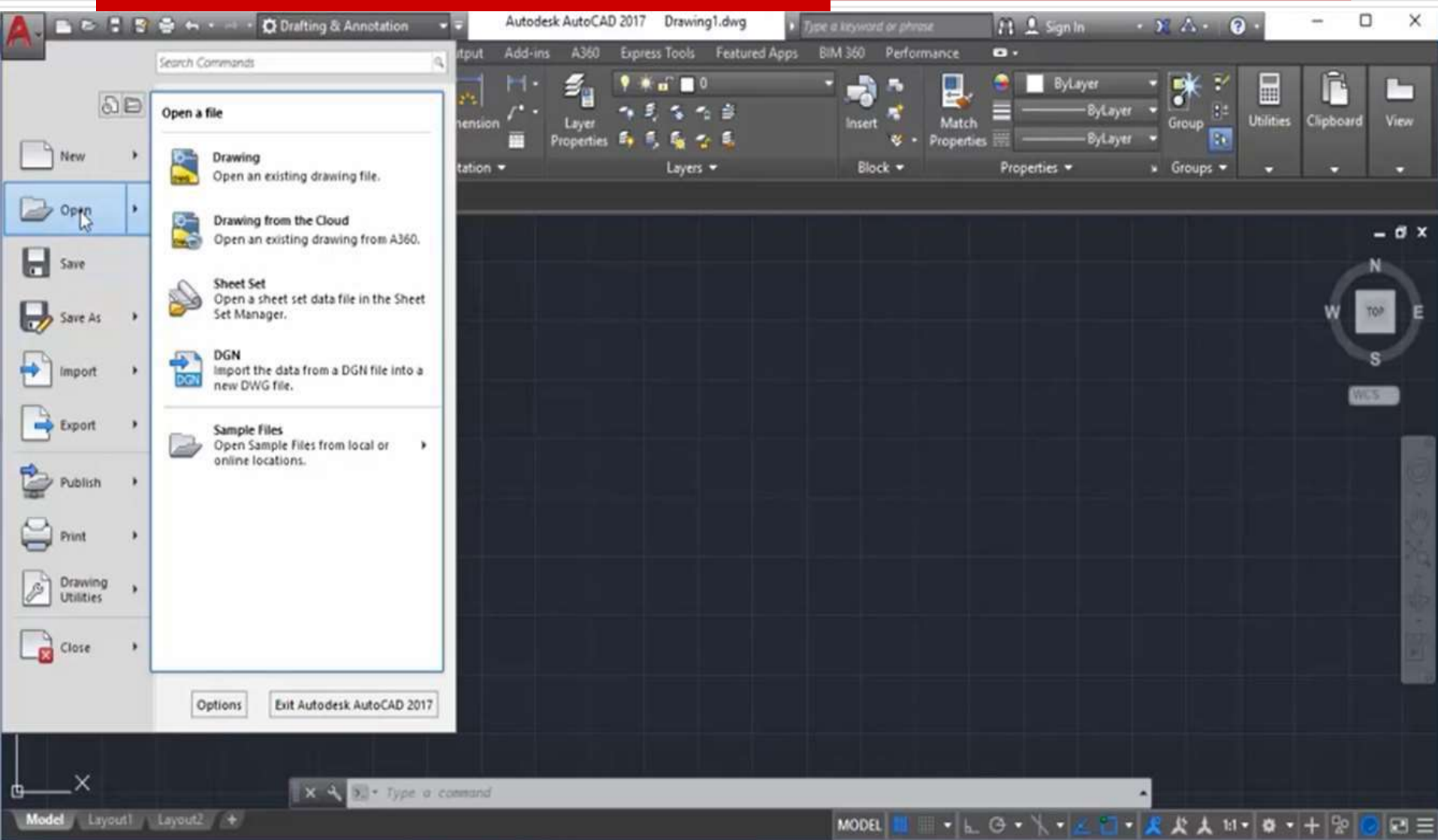
The Hitchhiker's Guide to AutoCAD Basics
Hitchhiker's AutoCAD Basics

LEARN **CREATE**

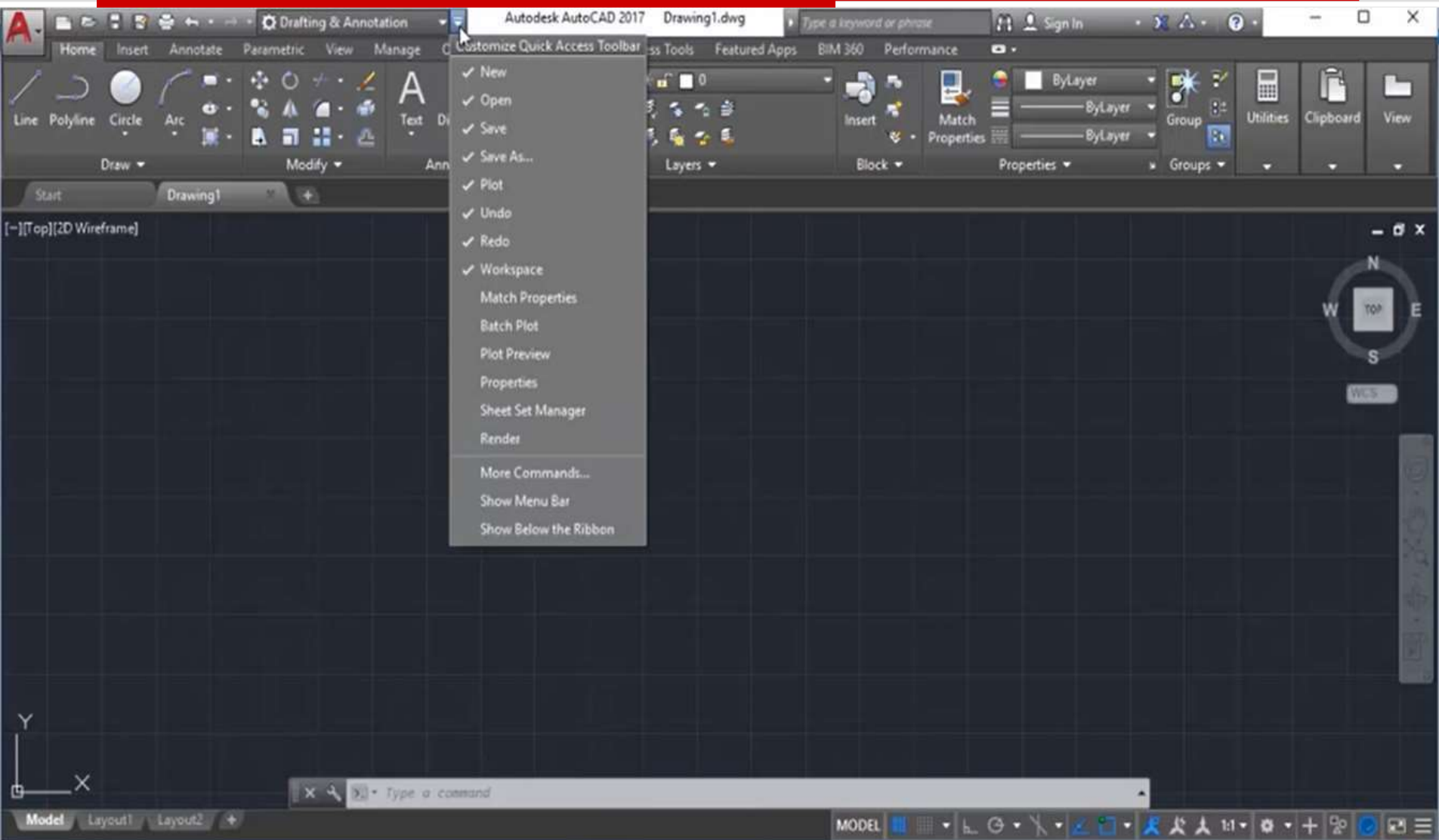
اجزاء واجهة AutoCAD؟



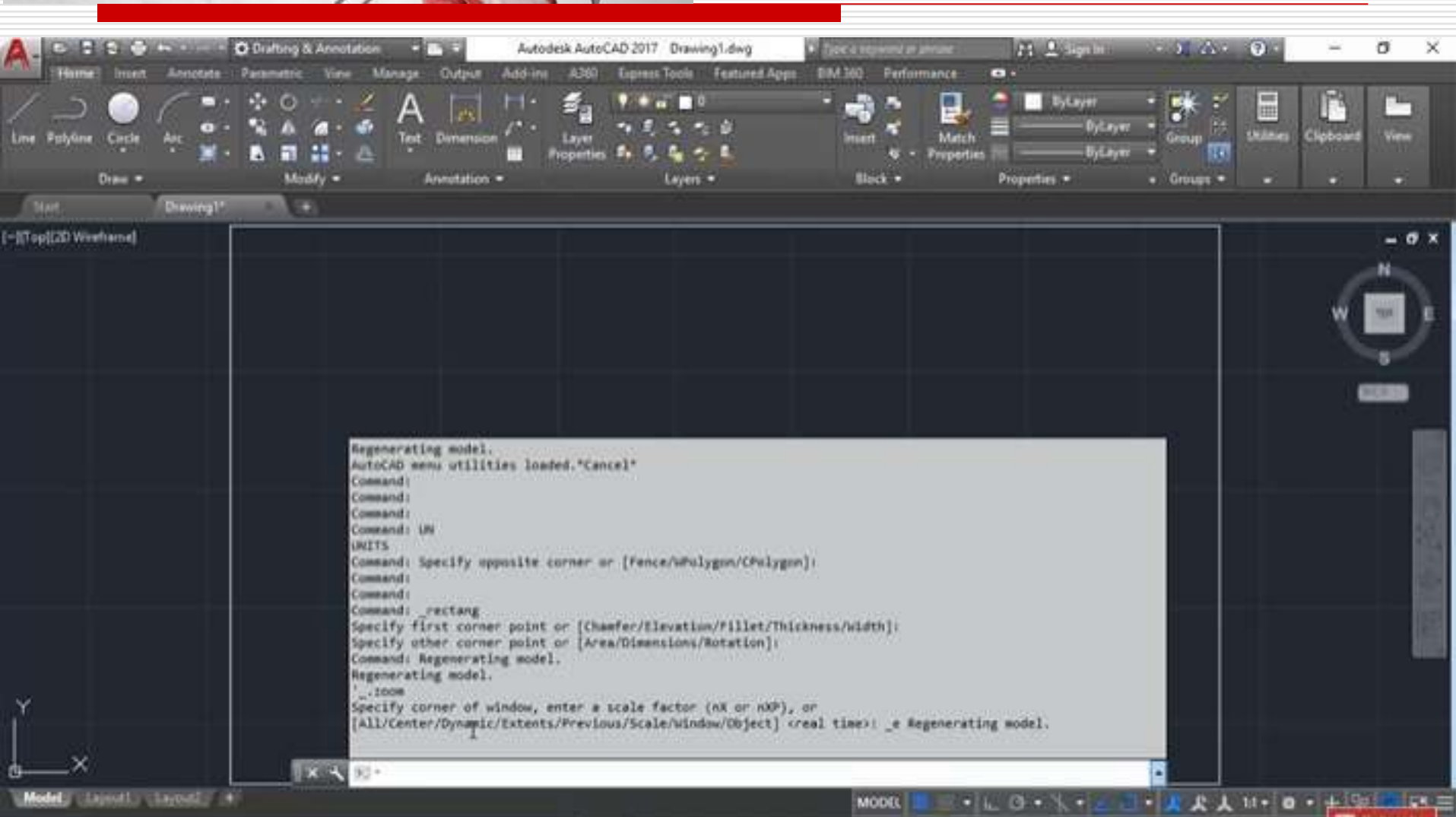
اجزاء واجهة AutoCAD؟



اجزاء واجهة AutoCAD؟



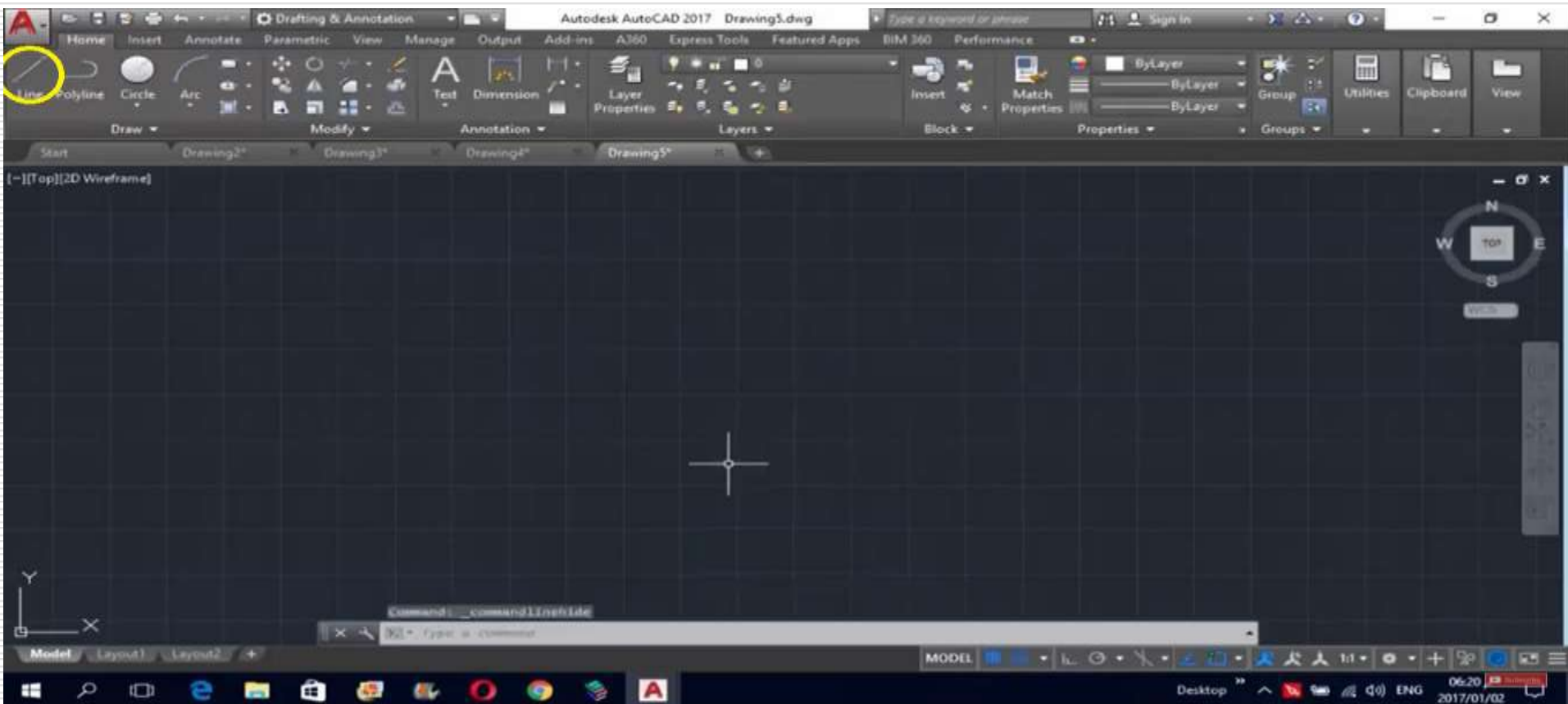
اجزاء واجهة AutoCAD؟



البدء مع AutoCAD؟

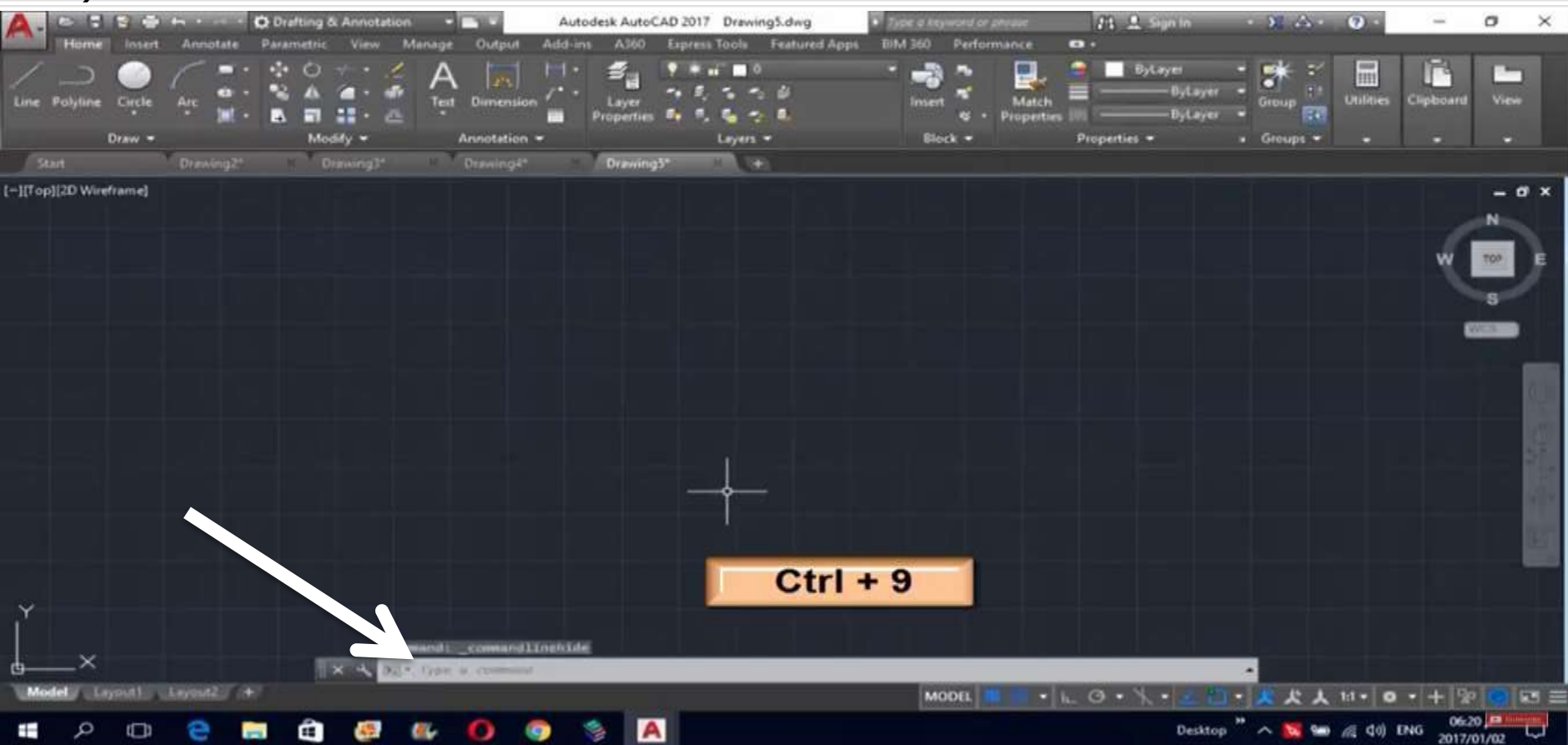
1- drawing a line

a) Home Ribbon + Mouse



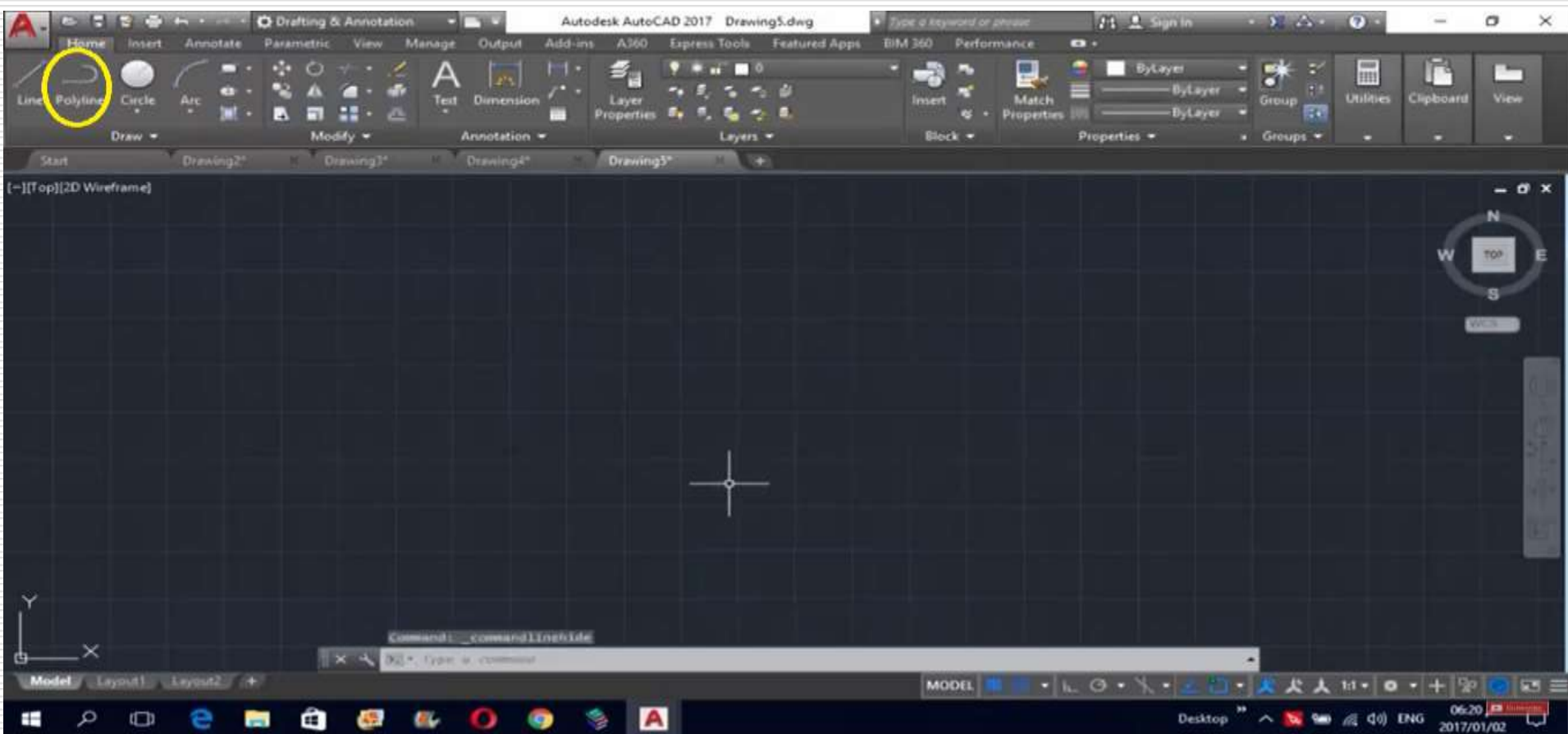
البدء مع AutoCAD؟

- 1- drawing a line
 - a) Command Line+Mouse



البدء مع AutoCAD؟

2- drawing a polyline



Lecture 2

الرسم الهندسي

أساسيات الرسم الهندسي

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محاضرة رقم (2)

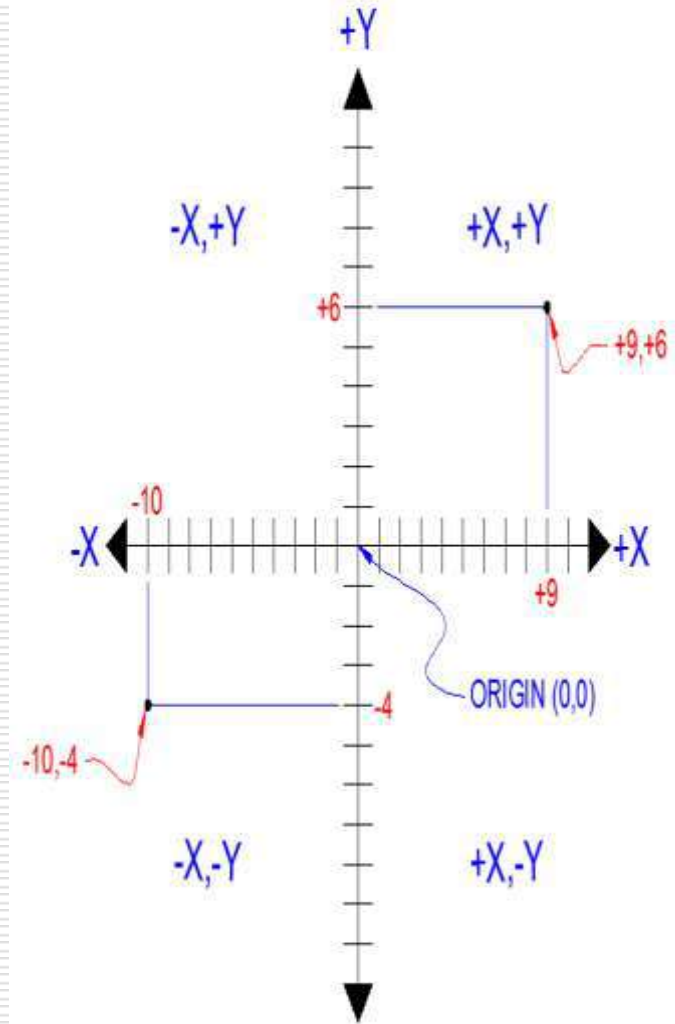
- يتعلم الطالب في هذه المحاضرة
- مراجعة لنظام الاحداثيات
- رسم النقاط والخطوط
- الاشكال ثنائية الابعاد وكيف نفرق بين كل منها (متوازي الاضلاع ، المستطيل ، المربع والدائرة)
- مدخل الى الخطوط (Line) وانواعها
- الخط المتعدد (Pline) والفرق بينه وبين مجموعة الخطوط



نظام الاحداثيات

➤ X,Y Coordinate System

- ❑ A point is the intersection of two coordinates X-axis , Y-axis
- ❑ A line has two points, a start point and an end point.
- ❑ AutoCAD uses points to determine where an object is located.
- ❑ The Origin point is (0,0) and all objects begin counting from it.

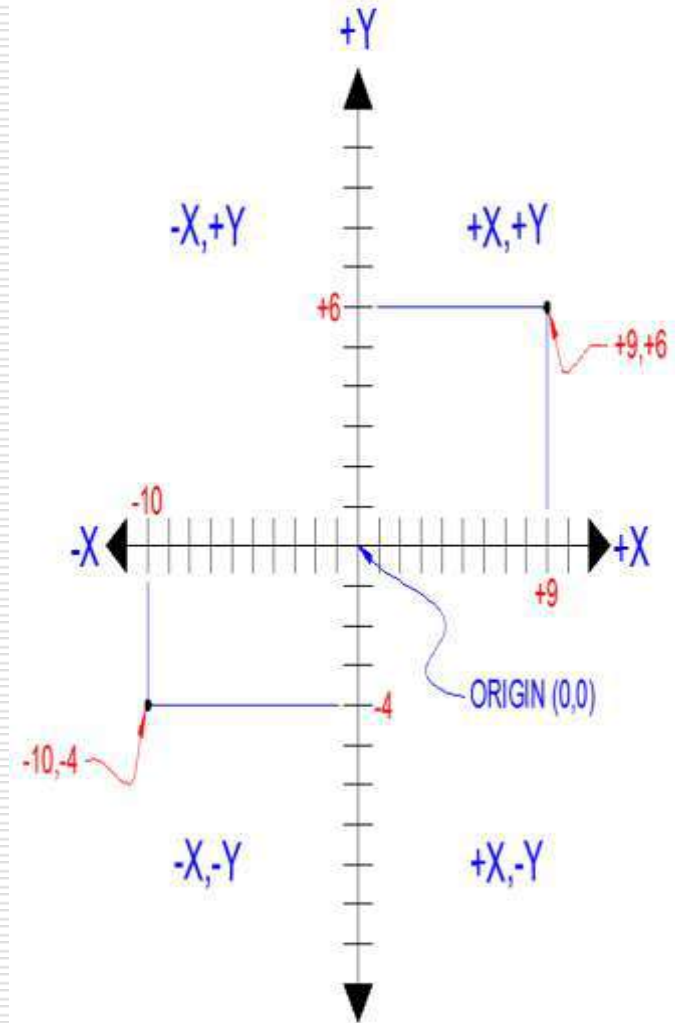




نظام الاحداثيات

➤ X,Y Coordinate System

- ❑ The positive X-axis is a line straight out to the right from the origin
- ❑ The positive Y-axis is a line straight up, this would be.
- ❑ The negative X-axis is a line straight out to the left from the origin
- ❑ The negative Y-axis is a line straight down.

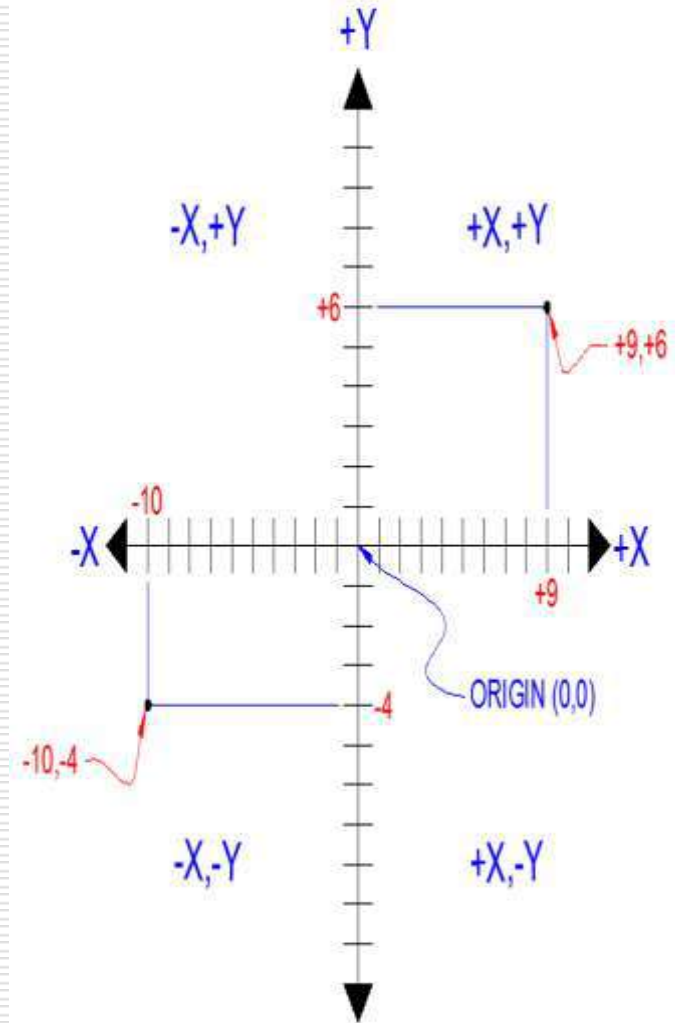




نظام الاحداثيات

➤ X,Y Coordinate System

- When you are working with points, X always comes first.
- Drawing (9,6) point going over +9 units over in the X-axis and +6 units up in the Y-axis.
- Drawing (-10,-4) point going 10 units in the negative X-axis (left) and 4 units in the negative Y-axis (down).



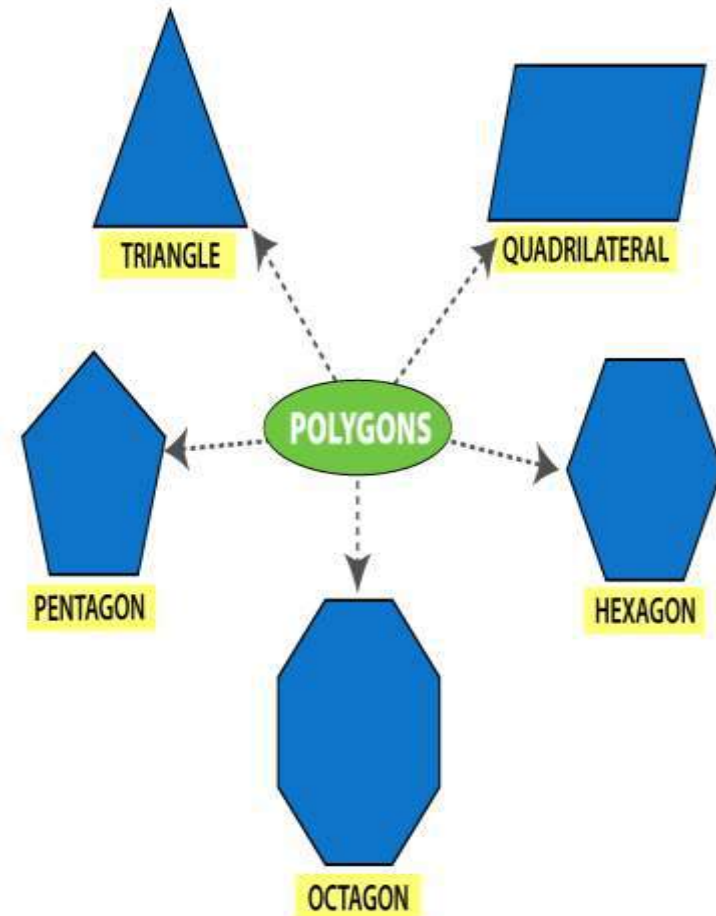


الاشكال ثنائية الابعاد

➤ B. 2D objects of lines and points

A polygon is a two-dimensional geometric figure that has a finite number of sides.

- ❖ Triangle (3 sides sum of angle = 180)
- ❖ Quadrilateral (4 sides sum of angle = 360)

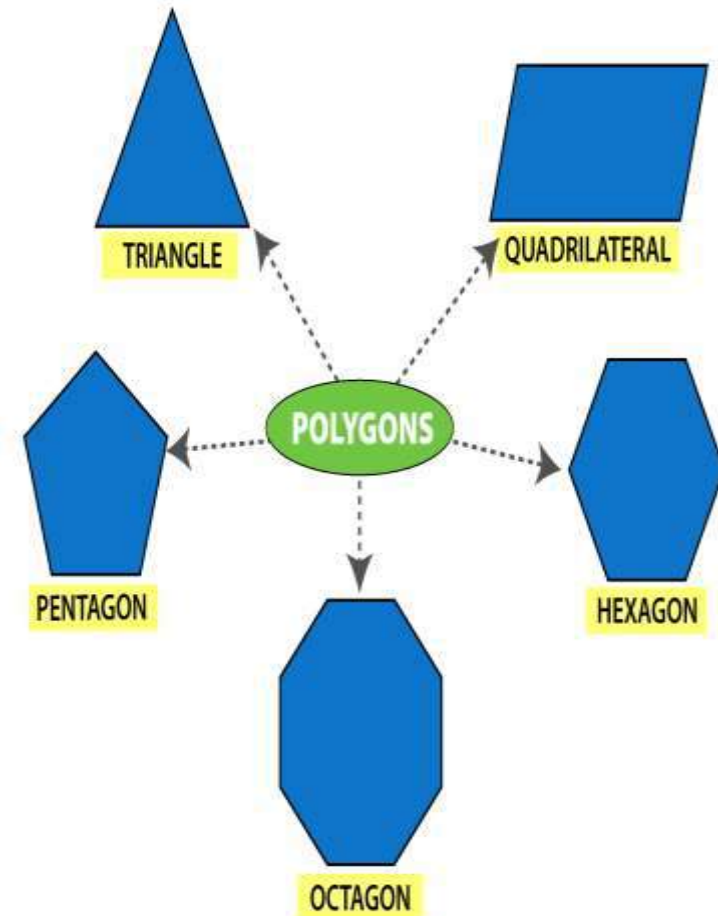




الاشكال ثنائية الابعاد

➤ B. 2D objects of lines and points

- ❖ Pentagon (5 sides sum of angle = 360)
- ❖ Hexagon (6 sides sum of angle = 360)
- ❖ Octagon (8 sides sum of angle = 360)



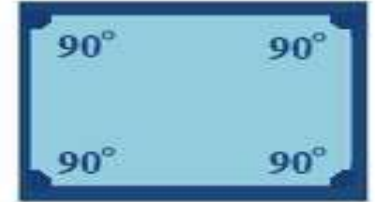


الاشكال ثنائية الابعاد

➤ B. 2D objects of lines and points

A quadrilateral shape can be divided into:

- ❖ Parallelogram: Each two opposite sides are equal and parallel, angles are not 90° . Opposite angles are equal
- ❖ Rectangular: Each two sides are equal and parallel, and all angles are 90° . Opposite angles are equal
- ❖ Square: Each two sides are parallel, all sides are equal, and all angles are 90° . Opposite angles are equal

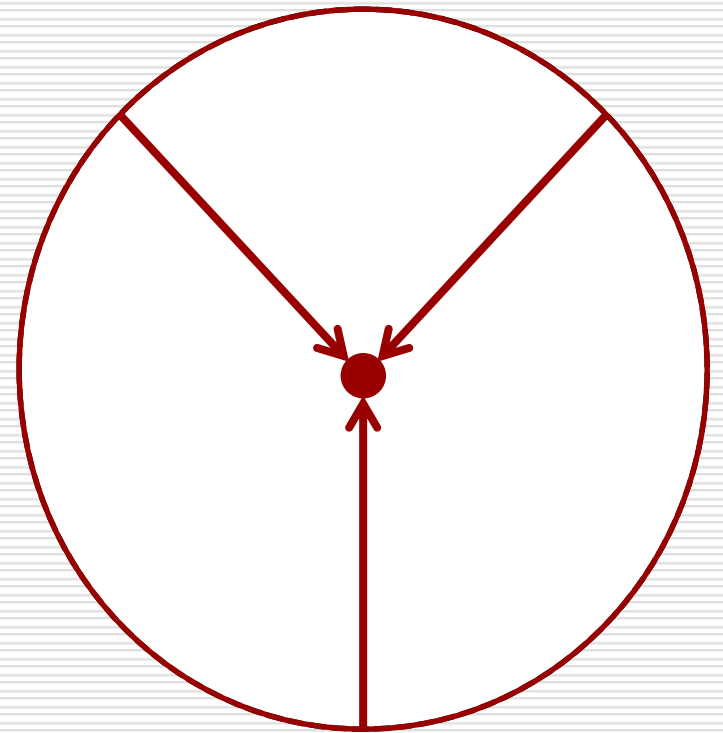




الاشكال ثنائية الابعاد

➤ B. 2D objects of lines and points

- ❖ Circle is:
- ❖ all points have the same distance from the center- OR
- ❖ set of lines between every two adjacent points





➤ C. Line Drawing

➤ A line is:

❖ In Mathematic:

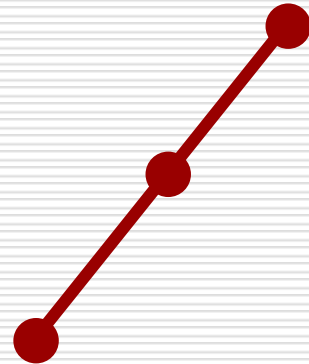
set of infinite points with a specific direction

❖ In Geometry:

The shortest way that connects between two points

❖ In CAD:

path that connects between the Begin and End points
according to the required path

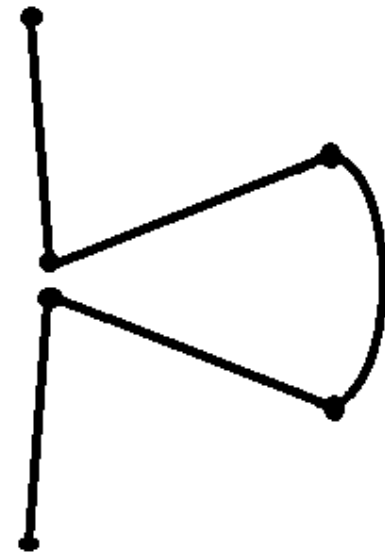




➤ C. Line Drawing

Polyline (Pline): is a procedure that allows more than one line of different types (vertical, horizontal, angle and arc) lines

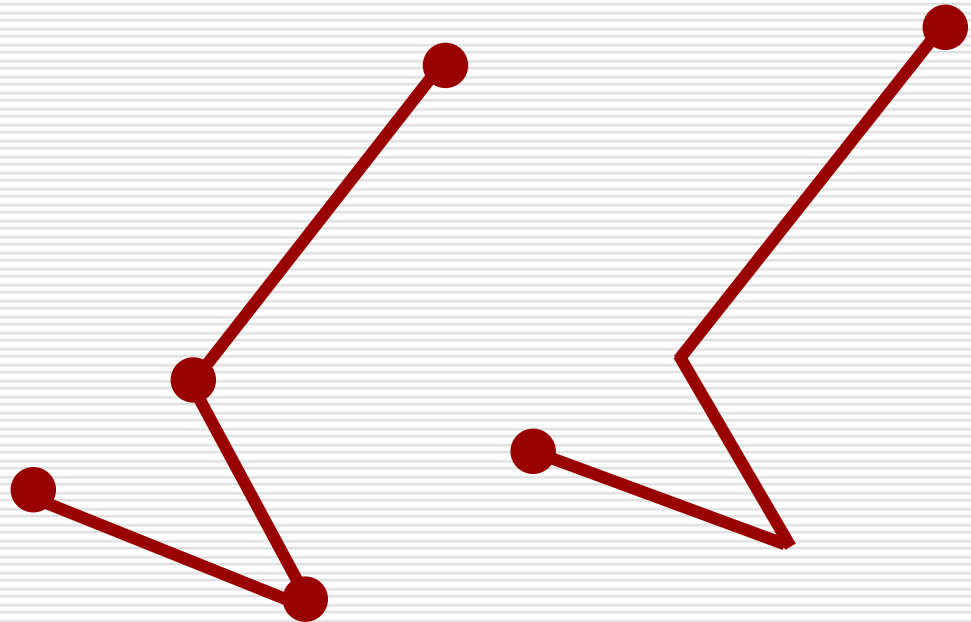
- ❖ Pline groups the all connected lines in one object.
- ❖ It's preferable when drawing set of lines as one part





➤ C. Line Drawing

The different between set of lines and Polyline is the grouping issue



Lecture 3

الرسم الهندسي

حالات ضبط الرسم الهندسي

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محاضرة رقم (3)

يتعلم الطالب في هذه المحاضرة

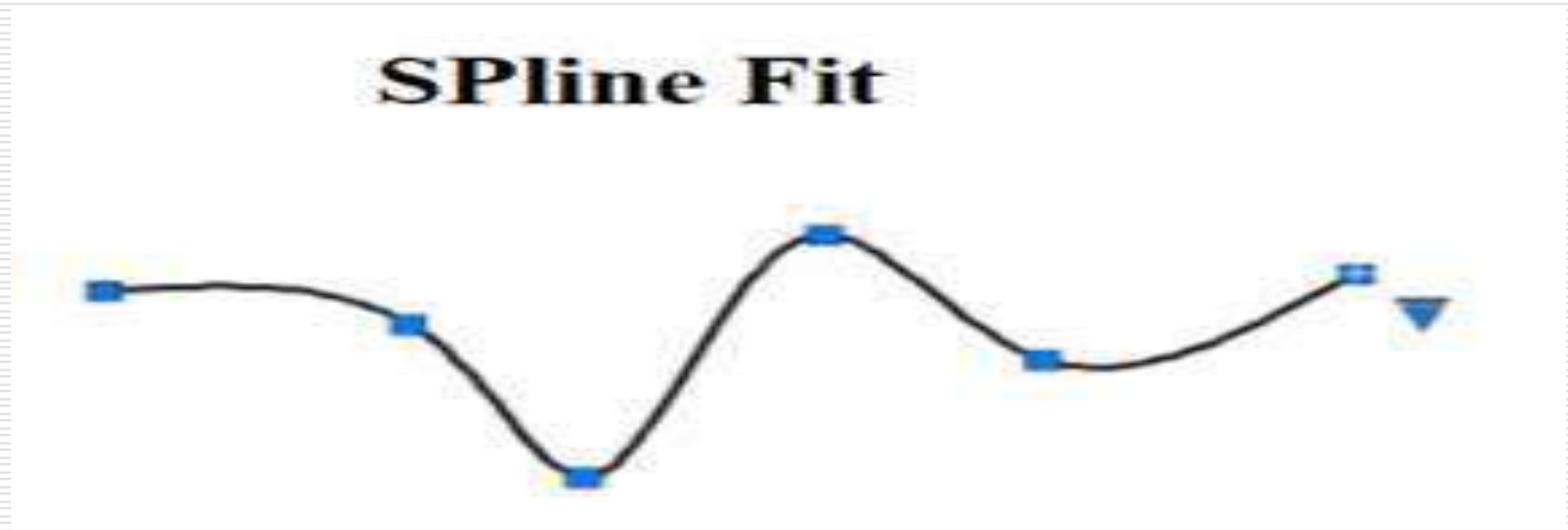
- الخط المتعدد (Smooth Poly Line- Spline)
- الخط (SP Line-Control Vertices- SP Line CV)
- Drawing settings



محاضرة رقم (3)

(Smooth Poly Line- Spline)

- ❖ **Spline Fit:** - To draw a spline which passes exactly through all the control points as shown in the below image.





محاضرة رقم (3)

(Smooth Poly Line- Spline)

- ❖ **Spline CV: - Also known as Control vertices method.** This option is to draw a spline that starts from the Starting Point and finish at the End Point and draw the best line that passes through (or near) the other control points

Spline CV

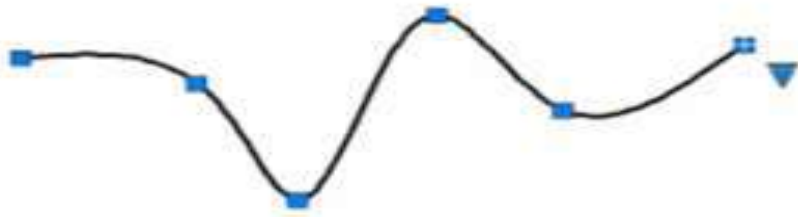




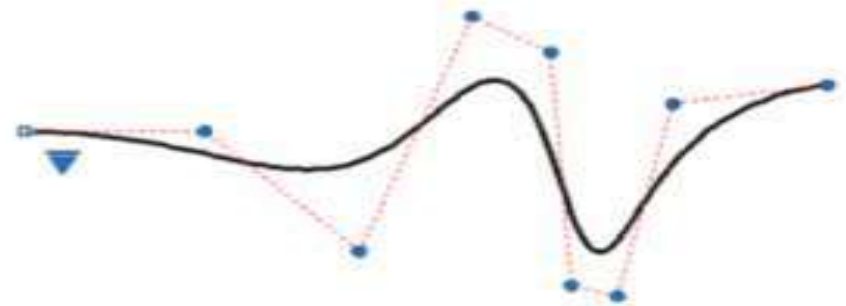
محاضرة رقم (3)

- ❖ Both the options are application dependent. Like few times you just need to connect the control points then use method of fit. If you are looking to fit your spline over the other image for the plot calculation you can go with CV method

Spline Fit



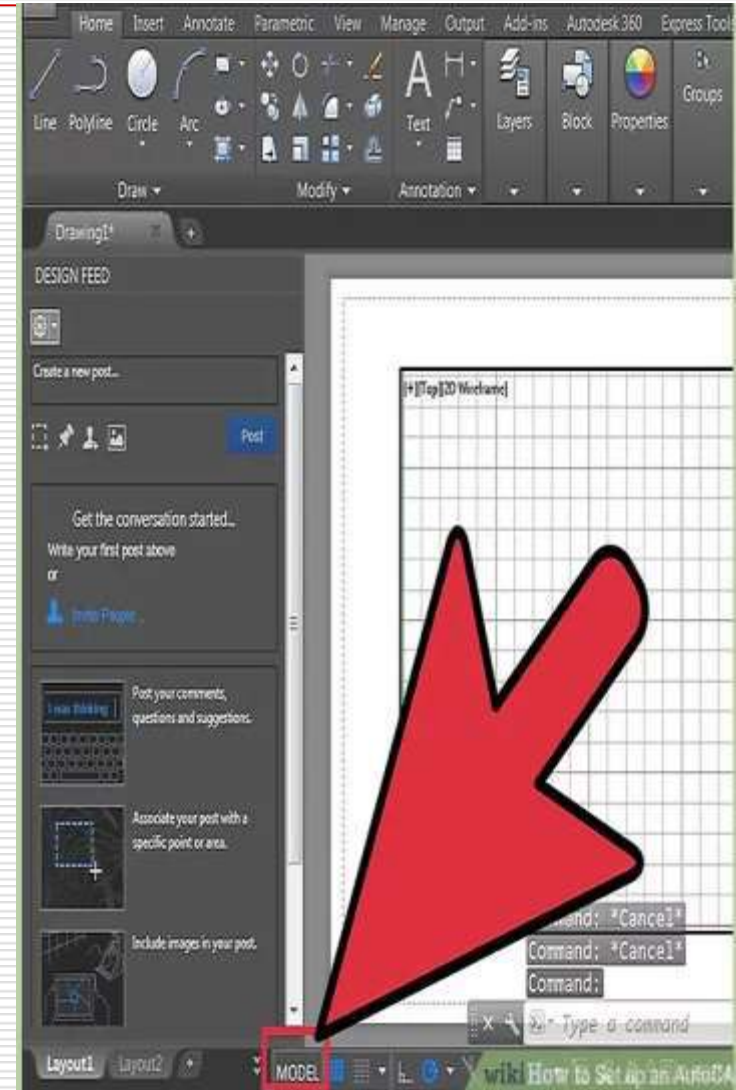
Spline CV





محاضرة رقم (3)

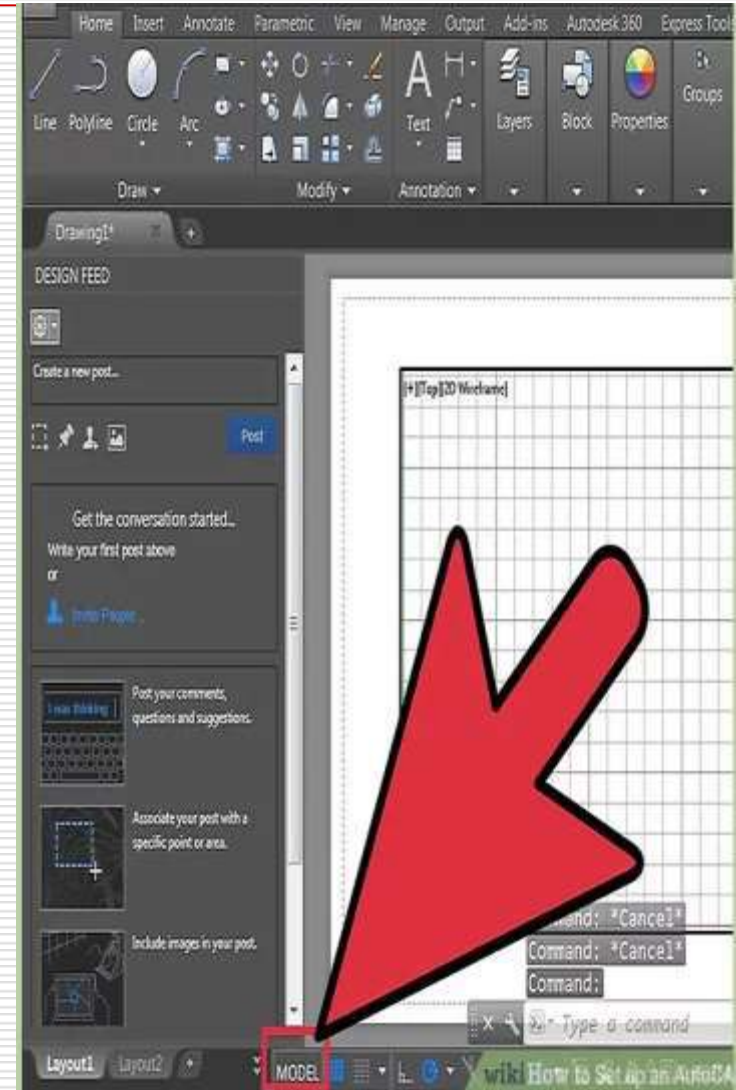
- 1) Open the AutoCAD program.
- 2) Go to model-space. There are two views in AutoCAD: model-space and paper-space. Your drawing should always be made in model-space, and dimensions that are added later should be represented in paper-space.





محاضرة رقم (3)

- 1) Open the AutoCAD program.
- 2) Go to model-space. There are two views in AutoCAD: model-space and paper-space. Your drawing should always be made in model-space, and dimensions that are added later should be represented in paper-space.





محاضرة رقم (3)

To switch between modelspace and paperspace, view the tabs located at the bottom of the screen. One tab is labeled 'Modelspace' and the other tabs are labeled either as 'Sheet' or 'Layout'. They indicate paperspace.

If you are in modelspace, the background of the screen should appear black. If you are in paperspace, the background should appear white.



محاضرة رقم (3)

3) Set Your Units.

There are different units : feet, meters, etc.

To ensure accuracy and eliminate confusion, it is important that the drawing is set in the proper units.

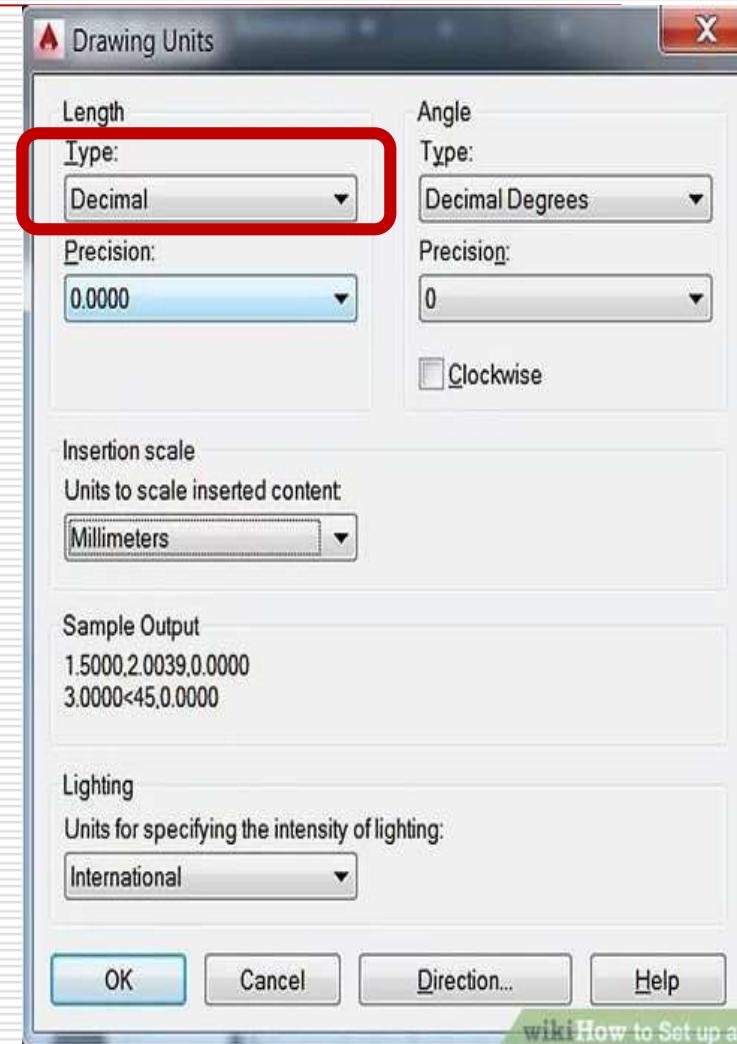
To set up your units, type 'UN' into your keyboard and then select the 'ENTER' key.



محاضرة رقم (3)

A dialog box should pop-up on the screen allowing you to specify the unit type and precision for your units.

The types are:
DECIMAL,
SCIENTIFIC, ENGINEERING,
ARCHITECTURAL,
FRACTIONAL.

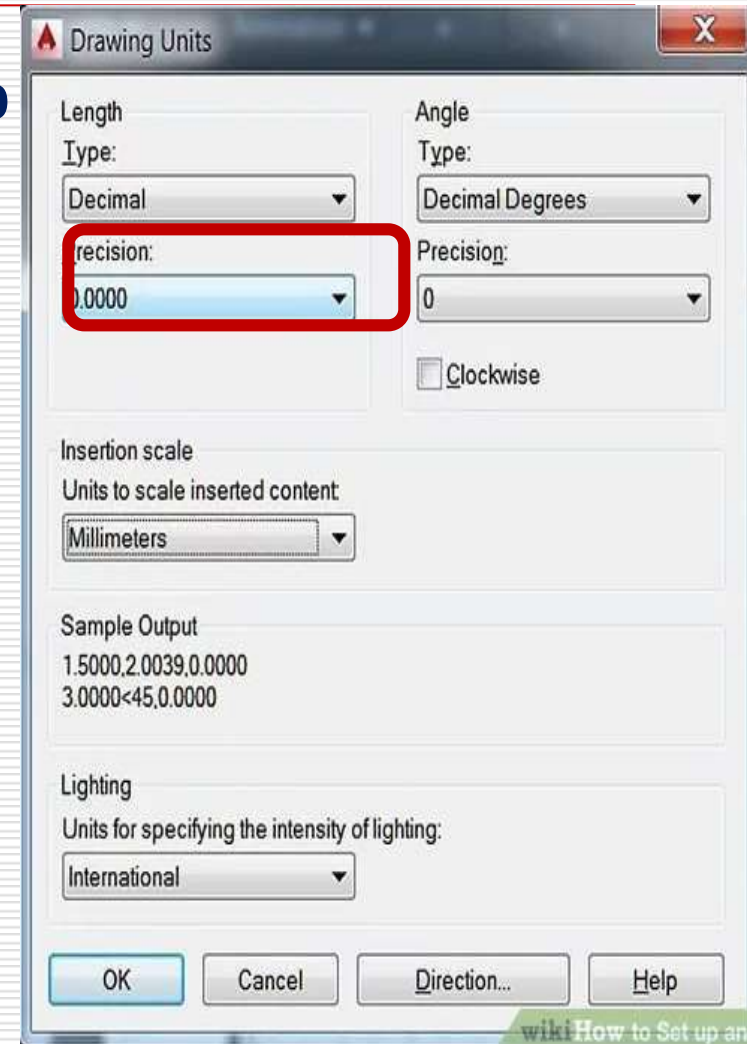




محاضرة رقم (3)

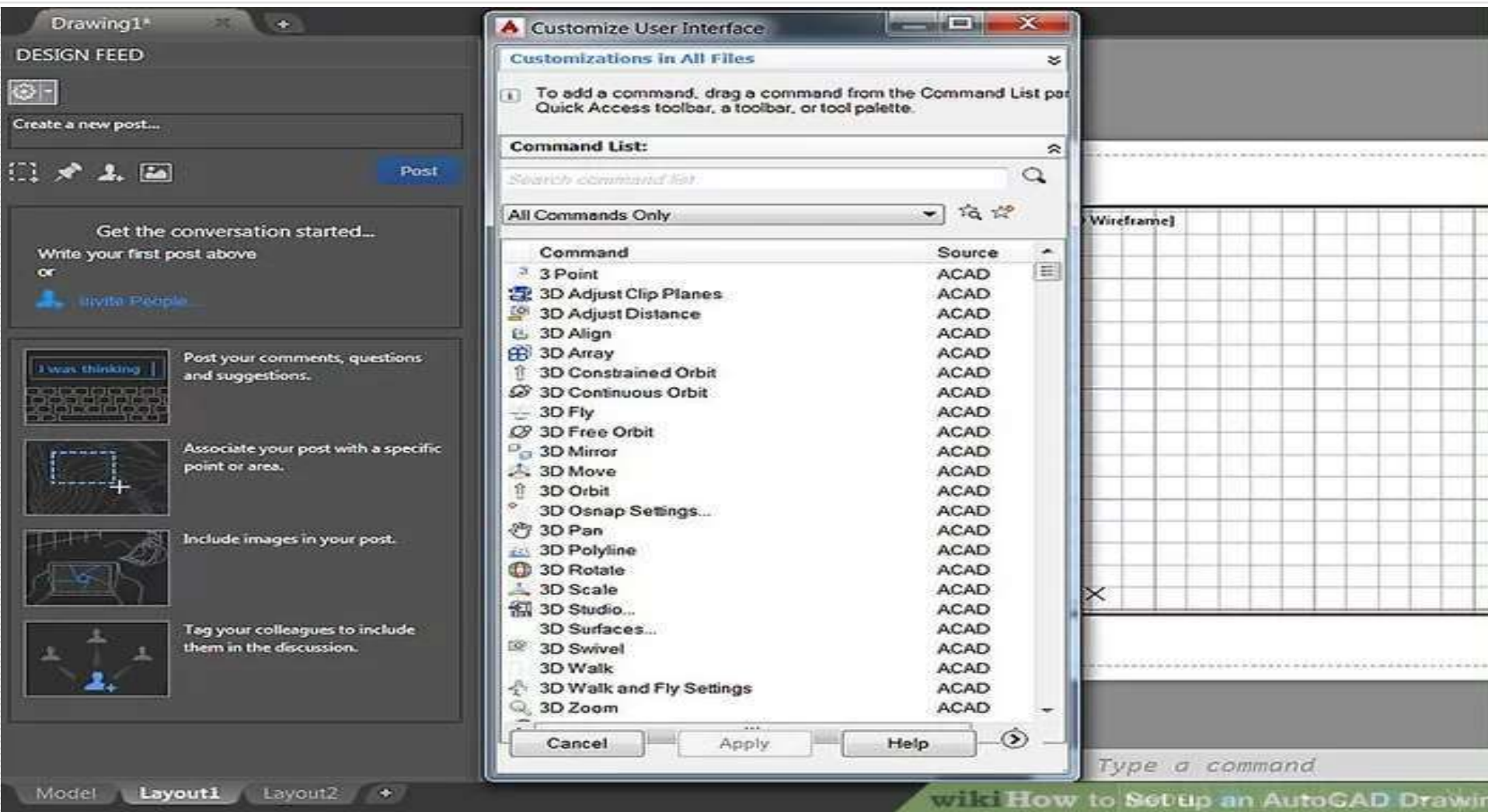
The 'precision' section allows you to pick the number of decimal places for your dimensions.

For any project, you should have information regarding unit specifications





محاضرة رقم (3)





محاضرة رقم (3)

The most popular toolbars that are used for AutoCAD 2D drawings are the

DRAW, MODIFY, and OBJECT PROPERTIES toolbars.

Select these toolbars and they should pop-up on your screen. Move them to the side to create space for your drawing.

DRAW TOOLBAR: contains common drawing tools.

MODIFY TOOLBAR: contains editing options.

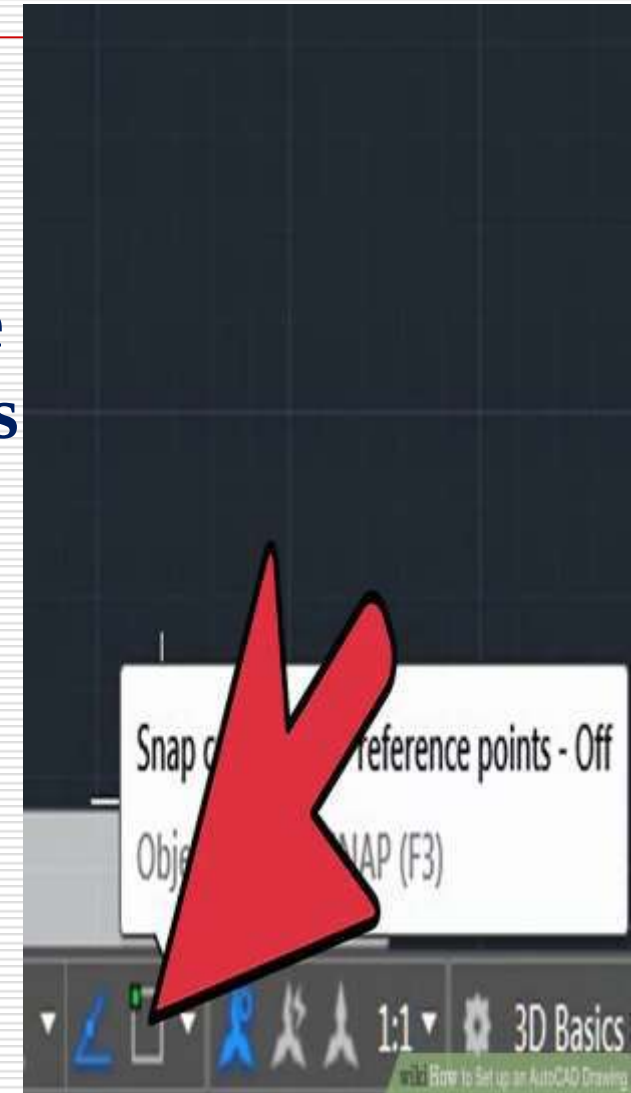
OBJECT PROPERTIES TOOLBAR: contains style and color options



محاضرة رقم (3)

5) Turn on OSNAP (Object Snap)

- ❖ it is a useful property when creating a drawing. It allows you to see where the midpoint and endpoint of a line is located, where a tangent is on a circle, and other useful information.
- ❖ To turn on OSNAP, press the F3 button on your keyboard.





محاضرة رقم (3)

5) Turn on OSNAP (Object Snap)

- ❖ To ensure your OSNAP settings are on, right-click on the 'OSNAP' icon located at the bottom left-hand corner of the screen.
- ❖ A dialog box should pop-up on the screen. Click the 'SELECT ALL' button, ensuring that all OSNAP properties are turned on.

Lecture 4

الرسم الهندسي

رسم الخط في الرسم الهندسي

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dr.ghalib@mtu.edu.iq



محاضرة رقم (4)

يتعلم الطالب في هذه المحاضرة

➤ خطوات رسم الخط

➤ تمارين على الخط



محاضرة رقم (4)

Line command enables you to draw straight line and there are 3 ways to draw the line

- 1.Go to Home → Draw → Line**
- 2.From the Draw Toolbar select line icon**
- 3.From the command line type the word Line (or just L) then press <Enter>**



محاضرة رقم (4)

Drawing a Line using Mouse

The first step is to specify the Start/ End points. This can be done by:-

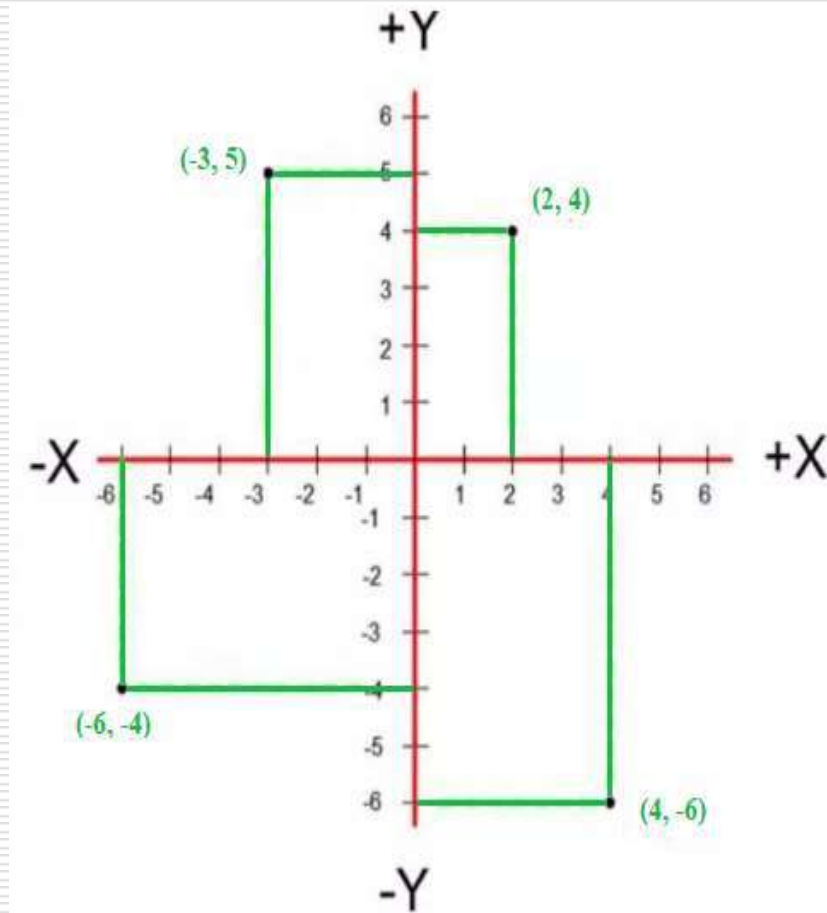
- (a) Move the cursor to Start point, press the left mouse-bottom, and move the mouse to the End point.**
- (b) Determine the point from the command line.**



محاضرة رقم (4)

Orthogonal Mode (Ortho)

If the Ortho mode is ON then the line will be horizontal or vertical only. If you want to other types of lines, convert the Ortho Mode to OFF case.

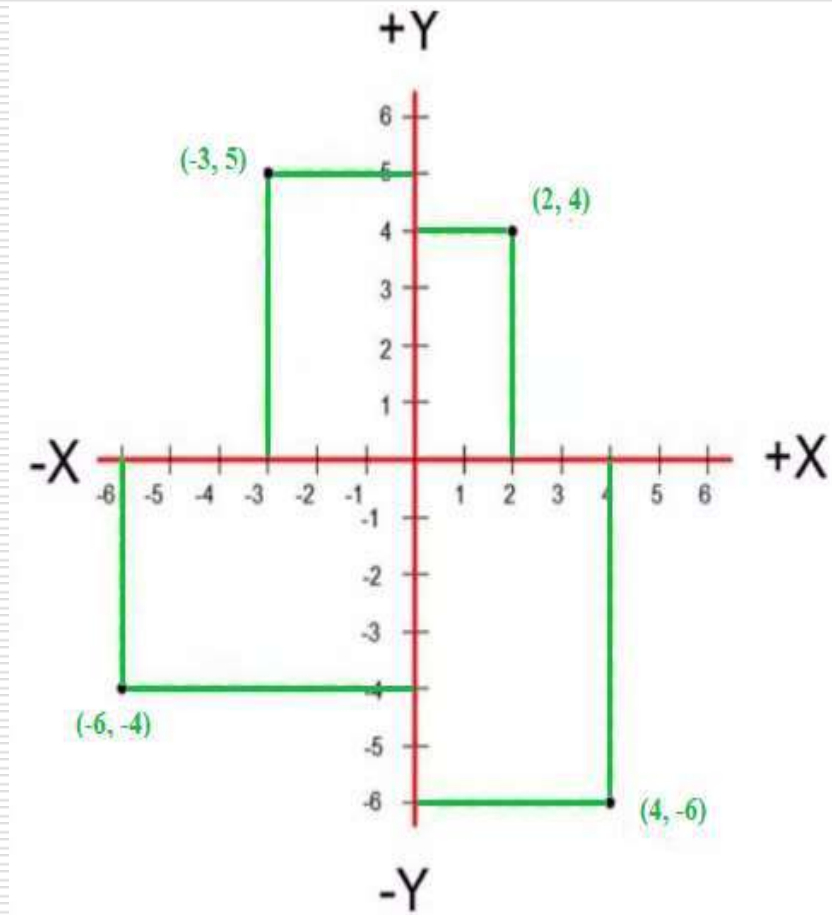




محاضرة رقم (4)

**To turn Ortho Mode On/ Off
press**

- F8 if you use Desktop computer
- Fn+F8 if you use Laptop

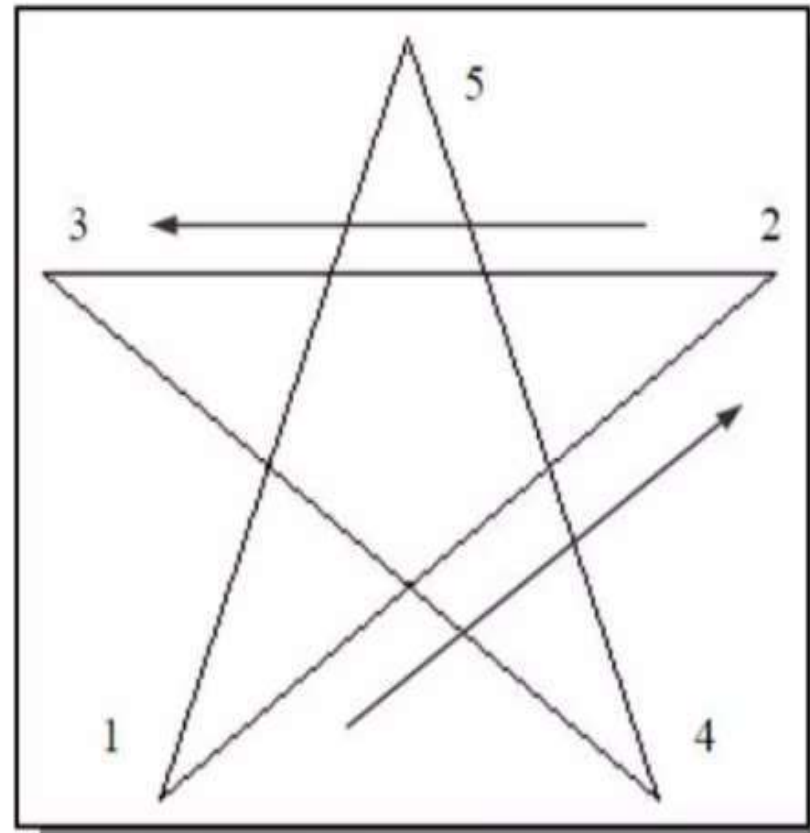




محاضرة رقم (4)

Exercise 1: Draw the next figure

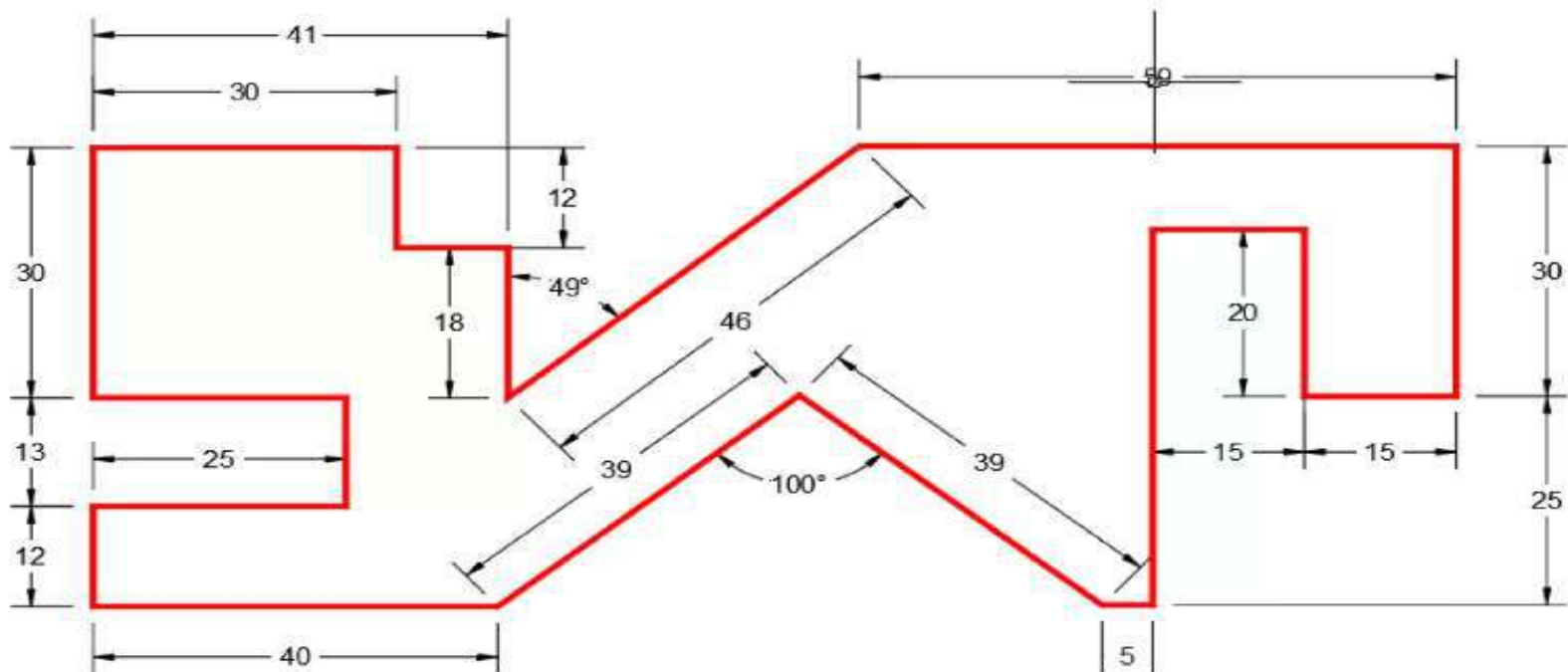
- Use command line NOT the mouse way
- Keep point order
- Size is NOT important.





محاضرة رقم (4)

Exercise 2: draw the next figure using command line and put the dimensions on it



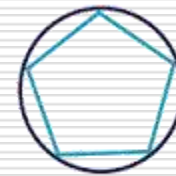


محاضرة رقم (4)

POLYGON

The polygon is a bounded region in a plane.

- It's drawn using set of points and lines.
- Each point between lines is a Vertex
- Each line between points is a Side



Inscribe in a circle



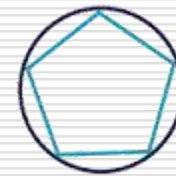
Circumscribe



محاضرة رقم (4)

POLYGON

To draw a polygon using command line follow the next steps



Inscribe in a circle



Circumscribe

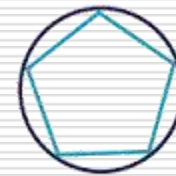
- Type POL on the command line and press Enter.
- Specify the number of sides. ...
- Press Enter. ...
- Specify the center of polygon on the viewport or workspace.



محاضرة رقم (4)

POLYGON

- Enter I or C for Inscribe and Circumscribe.
- Press Enter.
- Specify the radius of the polygon.



Inscribe in a circle

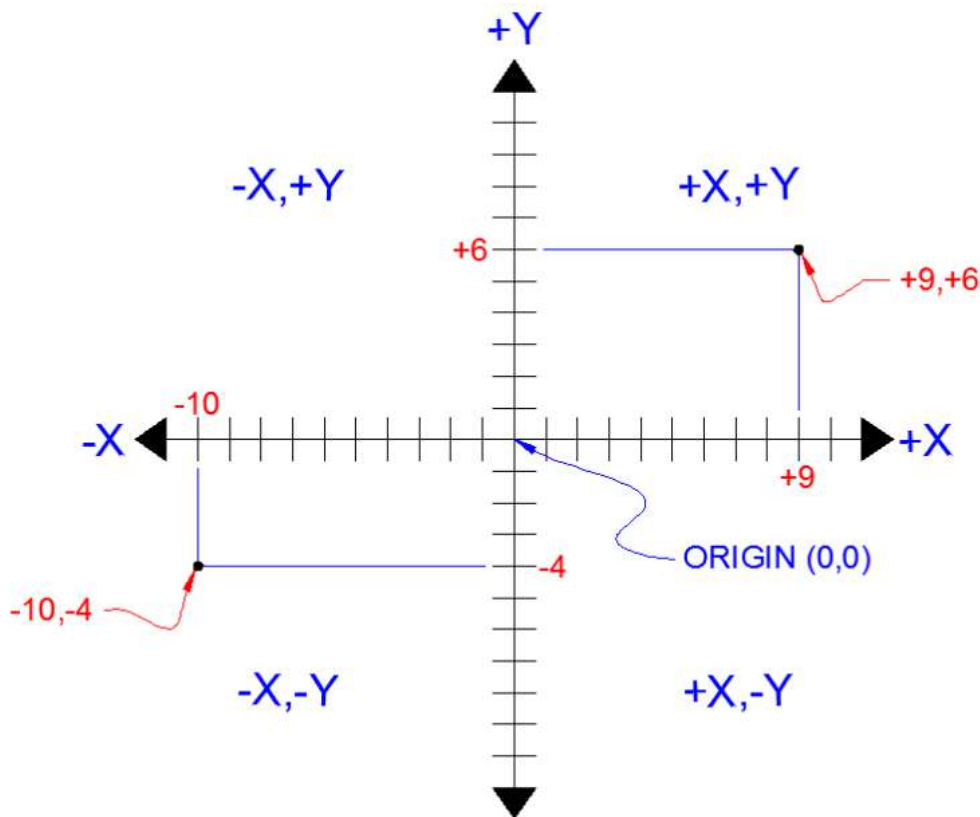


Circumscribe

1. The basic of AutoCAD

A. X,Y coordinate system

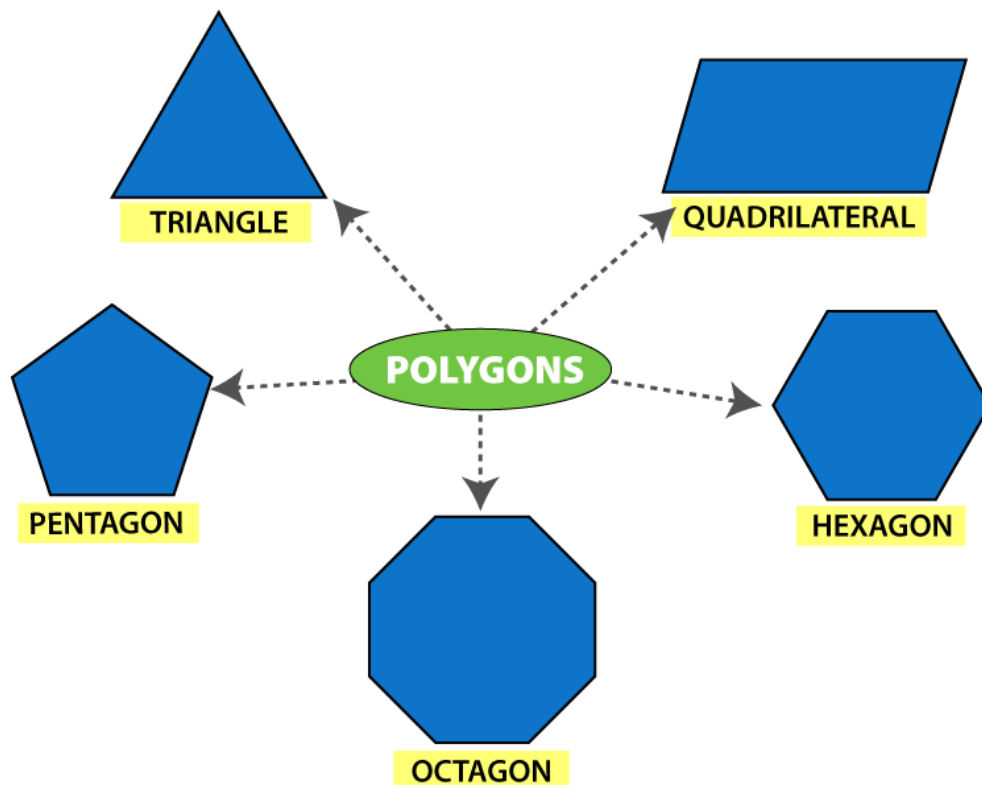
- ✓ A point is the intersection of two coordinates X-axis , Y-axis
- ✓ A line has two points, a start point and an end point.
- ✓ AutoCAD uses points to determine where an object is located.
- ✓ The Origin point is (0,0) and all objects begin counting from it.
- ✓ The positive X-axis is a line straight out to the right from the origin
- ✓ The positive Y-axis is a line straight up, this would be.
- ✓ The negative X-axis is a line straight out to the left from the origin



- ✓ The negative Y-axis is a line straight down.
- ✓ When you are working with points, X always comes first.
- ✓ Drawing a line to (9,6) point going over +9 units over in the X-axis and +6 units up in the Y-axis.
- ✓ The other point shown is (-10,-4). This means that the point is 10 units in the negative X-axis (left) and 4 units in the negative Y-axis (down).

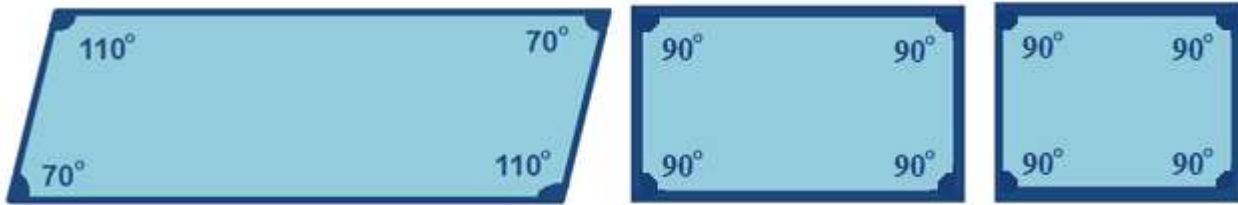
B. 2D objects of lines and points

- A **polygon** is a two-dimensional geometric figure that has a finite number of sides.
 - Triangle (3 sides sum of angle = 180)
 - Quadrilateral (4 sides sum of angle = 360)
 - Pentagon (5 sides sum of angle = 360)
 - Hexagon (6 sides sum of angle = 360)
 - Octagon (8 sides sum of angle = 360)
 - Circle (all points have the same distance from the center) OR (set of lines between every two adjacent points)

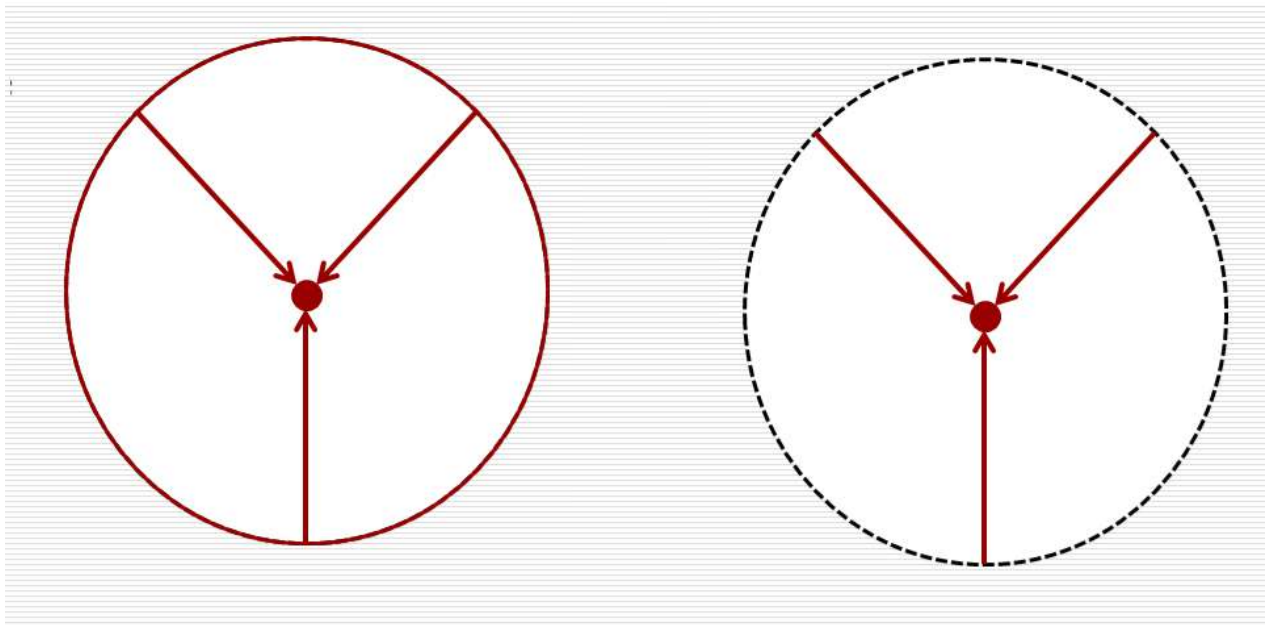


- A quadrilateral shape can be divided into:
 - ❖ Parallelogram: Each two opposite sides are equal and parallel, angles are not 90° . Opposite angles are equal
 - ❖ Rectangular: Each two sides are equal and parallel, and all angles are 90° . Opposite angles are equal
 - ❖ Square: Each two sides are parallel, all sides are equal, and all angles are 90° . Opposite angles are equal

➤ B. 2D objects of lines and points



- ❖ Circle is:
- ❖ all points have the same distance from the center- OR
- ❖ set of lines between every two adjacent points



Line Drawing

- **Line definition:** A line is:

- ❖ **In Mathematic:**

Set of infinite points with a specific direction

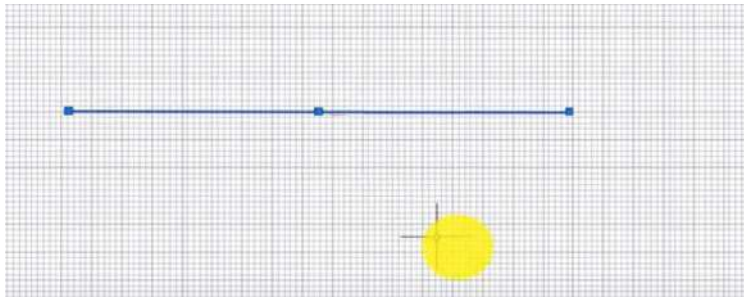
- ❖ **In Geometry:**

The shortest way that connects between two points

- ❖ **In CAD:**

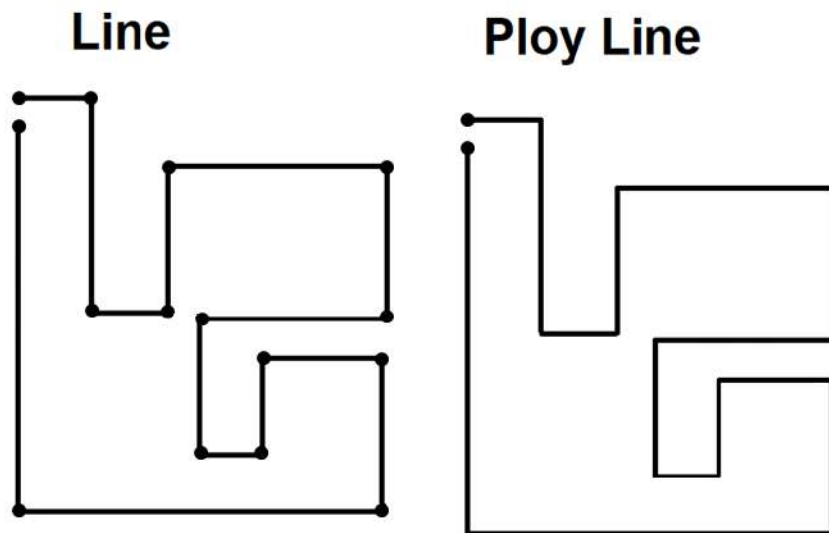
Path that connects between the Begin and End points according to the required path

- ✓ Vertical OR Horizontal (activate Orthomode)
- ✓ Any direction (deactivate Orthomode)



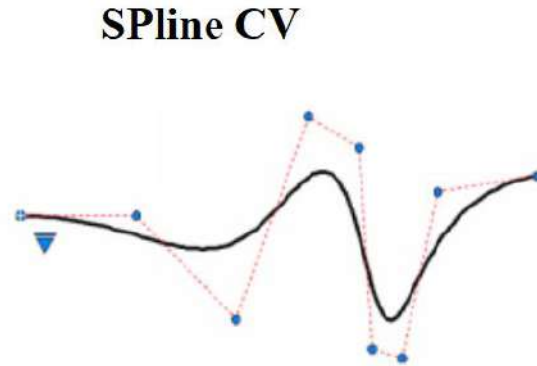
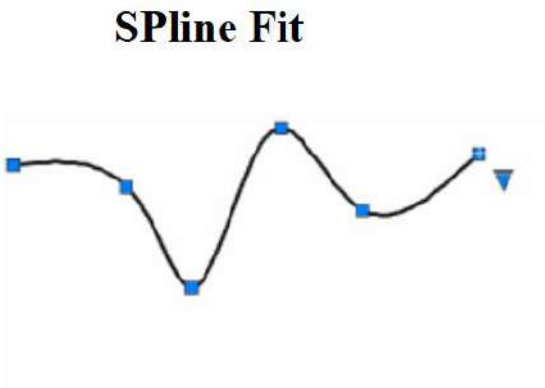
- ✓ **Poly Line (Pline):** is a procedure that allows more than one line of different types (vertical, horizontal, angle and arc) lines
- ❖ Pline groups the all connected lines in one object.
- ❖ It's preferable when drawing set of lines as one part

The different between set of lines and Polyline is the grouping issue



- ✓ **SP line**
 - Spline Fit: - To draw a spline which passes exactly **through all the control points** as shown in the below image.
 - Spline CV: - Also known as Control vertices method. This option is to draw a spline that starts from the Starting Point and finish at the

End Point and draw the best line that **passes through (or near) the other control points**

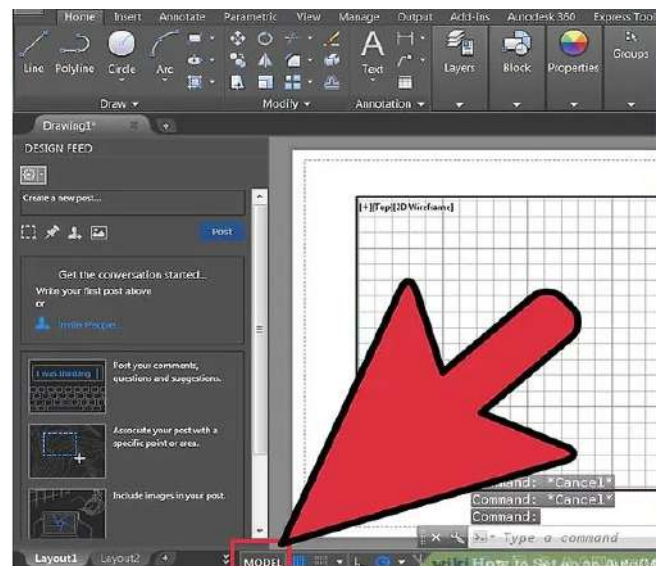


Both the options are application dependent. Like few times you just need to connect the control points then use method of fit. If you are looking to fit your spline over the other image for the plot calculation you can go with CV method

2. Steps of Drawing settings

- (a) Open the AutoCAD program. The program is either shown as an icon on your desktop, or you may find it in the START menu on the bottom left corner of your computer screen
- (b) Go to model-space. There are two views in AutoCAD: model-space and paper-space. Your drawing should always be made in model-space, and dimensions that are added later should be represented in paper-space.

To switch between model-space and paper-space, view the tabs located at the bottom of the screen. One tab is labeled 'model-space' and the other tabs are labeled either as 'sheet' or 'layout'. The 'sheet' or 'layout' tabs indicate paper-space. If you are in model-space, the background of the

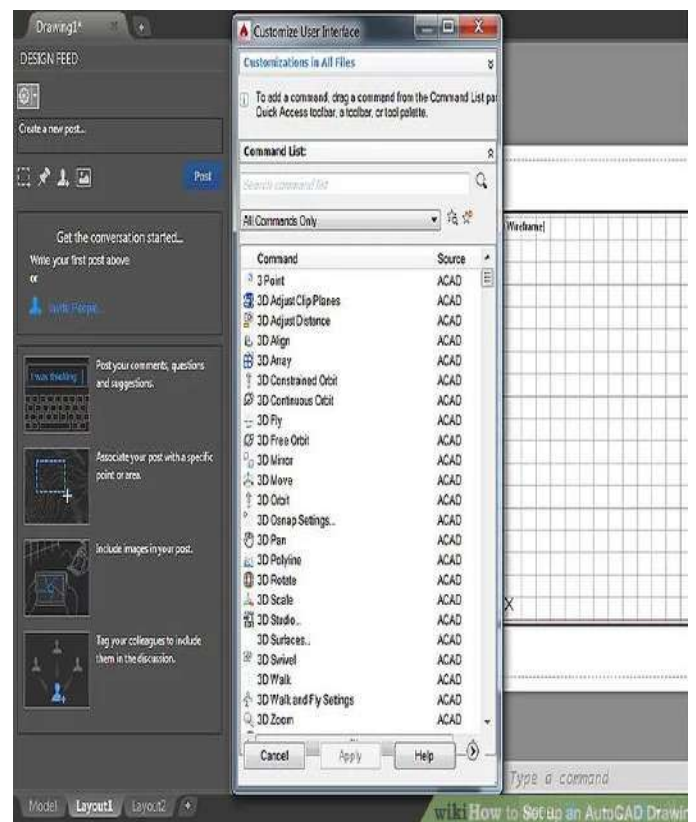


screen should appear black. If you are in paper-space, the background should appear white.

- (c) Set your units. Engineers represent units in various ways: feet, meters, etc. To ensure accuracy and eliminate confusion, it is important that the drawing is set in the proper units. To set up your units, type 'UN' into your keyboard and then select the 'ENTER' key. A dialog box should pop-up on the screen allowing you to specify the unit type and precision for your units. The options for the unit types are: Decimal, Scientific, Engineering, Architectural, and Fractional. The 'precision' section allows you to pick the number of decimal places for your dimensions. If you are doing a project for your teacher, he should have information regarding unit specifications



- (d) Select the toolbars that you will use throughout your drawing. To do this, put your mouse over an empty space at the top of your screen near the toolbars. Then right-click and select AutoCAD. A long list should show up, displaying various toolbars containing different commands. The most popular toolbars that are used for AutoCAD 2D drawings are the DRAW, MODIFY, and OBJECT PROPERTIES toolbars. Select these toolbars and they should pop-up on your screen. Move them to the side to create space for your drawing. DRAW TOOLBAR: contains common drawing tools. MODIFY TOOLBAR: contains editing options. OBJECT PROPERTIES



TOOLBAR: contains style and color options.

- (e) Turn on OSNAP. OSNAP, which refers to object snap, is an extremely useful property when creating a drawing. It allows you to see where the midpoint and endpoint of a line is located, where a tangent is on a circle, and other useful information. To turn on OSNAP, press the F3 button on your keyboard. To ensure your OSNAP settings are on, right-click on the icon reading 'OSNAP' located at the bottom left-hand corner of the screen. A dialog box should pop-up on the screen. Click the 'SELECT ALL' button, ensuring that all OSNAP properties are turned on.



3. Drawing the Line

Line command enables you to draw straight line and there are 3 ways to draw the line

1. Go to Home → Draw → Line
2. From the Draw Toolbar select line icon
3. From the command line type the word Line (or just L) then press <Enter>

The first step in drawing a line is to specify the first point. This can be done by:-

- (a) Move the cursor to **Start** point, press the left mouse-bottom, and move the mouse to the **End** point.
- (b) Determine the point from the command line.

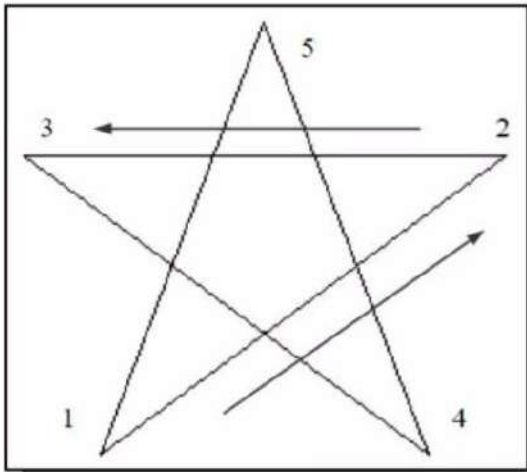
Ortho Mode: if the ortho mode is ON then the line will be horizontal or vertical only. If you want to other types of lines, convert the Ortho Mode to OFF case.

To turn Ortho Mode On/ Off press

- F8 if you use Desktop computer
- Fn+F8 if you use Laptop

Exercise 1: Draw the next figure

- use command line NOT the mouse way
- Keep point order

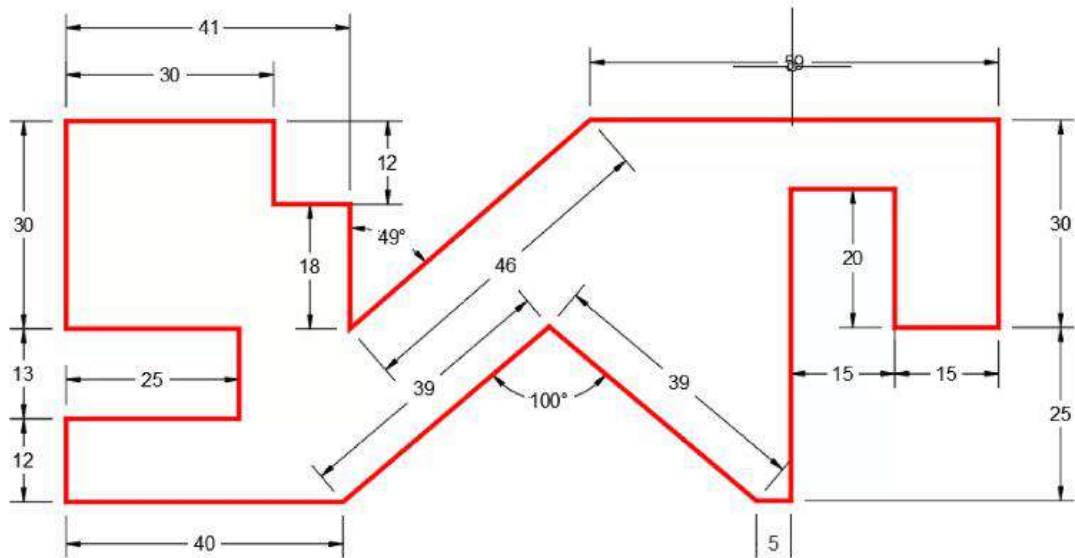


تمرين: ارسم الشكل التالي مع ملاحظة

- لا تستخدم الماوس للرسم وانما سطر الاوامر (Command Line)
- حافظ على ترتيب النقاط رؤوس الرسم حسب الشكل
- الحجم غير مهم

- DONOT pay attention for the size.

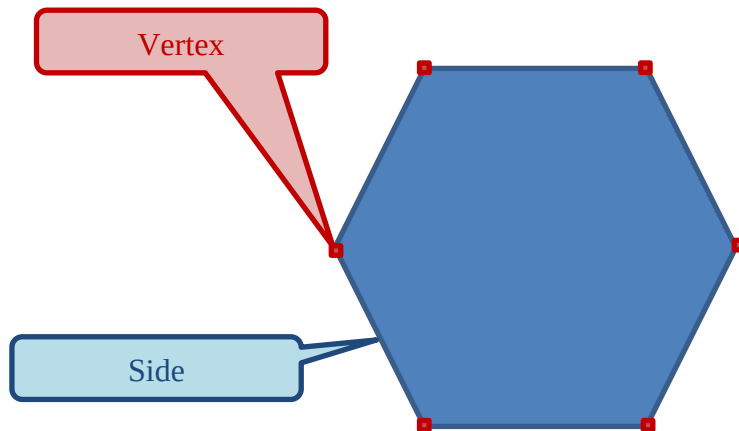
Exercise 2: draw the next figure using command line and put the dimensions on it



4. Drawing the Polygon

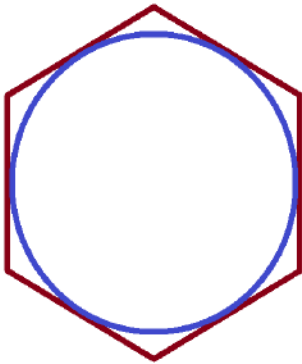
The **polygon** is a bounded region in a plane.

- It's drawn using a set of points and lines.
- Each point between lines is a **Vertex**
- Each line between points is a **Side**

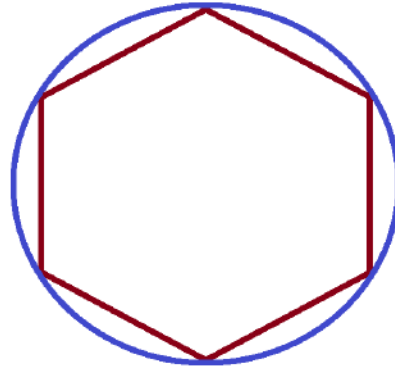


To draw a polygon using command line, follow the next steps:

- Type POL on the command line and press Enter.
- Specify the number of sides. ..Press Enter. ...
- Specify the center of polygon on the viewport or workspace.
- Specify the radius of the polygon.
- Enter I for Inscribe or C for Circumscribe.... Press Enter.



Inscribe in the Circle



Circumscribe outside the Circle